

Abstract

Here we introduce a complete pipeline to extract co-expressed gene cluster (metagene) from TS19 mouse embryo, including the biological insights of these metagenes. Also, we provide several examples of using metagenes to understand gene functions.

Inspired by cell-oriented studies described in previous chapters and related work in gene co-expression network, we find that by transposing the expression matrix thus consider genes as samples while cells as variables, we are able to project genes onto reduced dimensional space just as what we did for cells. Compared to canonical approaches, we are able to extract gene clusters with specific biological functions. This will provide a systematic approach which helps understand gene functions.

Introduction

Gene co-expression network

A key objective of system biology research is to systematically understand gene functions and how different genes interact. However, until recently our understanding of the genome is still very limited. Of all protein coding genes in mouse, about half have no experimental based annotations. For the annotated genes, many of which have only a few GO terms related (Figure and REF), indicating lack of insights of gene functions. Having a systematic approach that helps predicting gene function is crucial to understand our genome [and other significance].

One of the attempts scientists have tried is studying patterns of gene expression. Back in 1998, scientists confirmed that by clustering gene expression data from microarray, genes of similar function could be clustered together(Eisen et al. 1998). After that, various approaches based on gene co-expression have been developed to study conserved genetic modules(Stuart et al. 2003), find biological relevant genes to predict gene function(Butte and Kohane 2000; Luo et al. 2007; van Dam et al. 2018), characterize cellular state(Carter et al. 2004) and study gene evolution(Ruprecht et al. 2017). A human RNA-seq-based gene co-expression database has also been established(van Dam, Craig, and de Magalhães 2015). However, these methods have certain limitations in their ability to capture full structure of the data, also the bulk RNA seq based expression level has been averaged across all sample cells thus considered as low resolution. Nowadays, with scRNA seq technology, it is possible to overcome these hurdles by obtaining sequencing data with larger scale and higher resolution.

Cell type specific gene and housekeeping gene

To be added

Cell cycle regulation

In eukaryotes, cell cycle is directly related to several important biological processes, involving DNA replication, chromosome segregation and cell growth. Typically, eukaryotic cell cycle can be divided into four phases: G1 phase, S phase, G2 phase and M phase. In a living organism most of cells are at G1 phase, during which cells increase their size and prepare for DNA synthesis(Lemons et al. 2010). At S phase, DNA replication occurs. G2 phase is the gap between S (synthesis) phase and M (mitosis) phase, where cells

continue to grow and prepare for mitosis. At M phase, mitosis takes place and cells get divided into two daughter cells. Through the past few decades, scientists revealed several dependencies that control cell cycle(Barnum and O'Connell 2014), such as cell size(Mitchison and Nurse 1985; Mitchison 1957; Fantes 1977), DNA damage(Weinert and Hartwell 1988; Hartwell and Weinert 1989), completion of DNA replication(Matthew J. O'Connell and Cimprich 2005; M. J. O'Connell, Walworth, and Carr 2000; Kuntz and O'Connell 2009; Tapia-Alveal, Calonge, and O'Connell 2009), microtubule formation(Li and Murray 1991; Hoyt, Totis, and Roberts 1991) and chromosome segregation(Lara-Gonzalez, Westhorpe, and Taylor 2012).

Result

1. Setup metagene extraction workflow based on canonical scRNA seq pipeline

The frame of metagene extraction is similar to cell cluster extraction from canonical scRNA seq pipeline, including the following parts:

- I. Data normalization
- II. Dimension reduction by PCA and tSNE
- III. Clustering
- IV. Biological interpretations

Based on this framework, we start with a transposed expression matrix with genes as 'samples' and cells as 'variables', then a few modifications are made to achieve better quality metagenes. In our preliminary test to evaluate different variables in step I and II, metagenes are extracted by Ward Hierarchical clustering of resulted tSNE coordinates, GSEA is then applied to all extracted metagenes and number of matched GO terms are

compared across different variables we tested. The gene-gene correlation of three sets of marker genes are used as benchmark:

- a). 76 ribosomal protein genes
- b). G1 phase marker genes
- c). G2/M phase marker genes

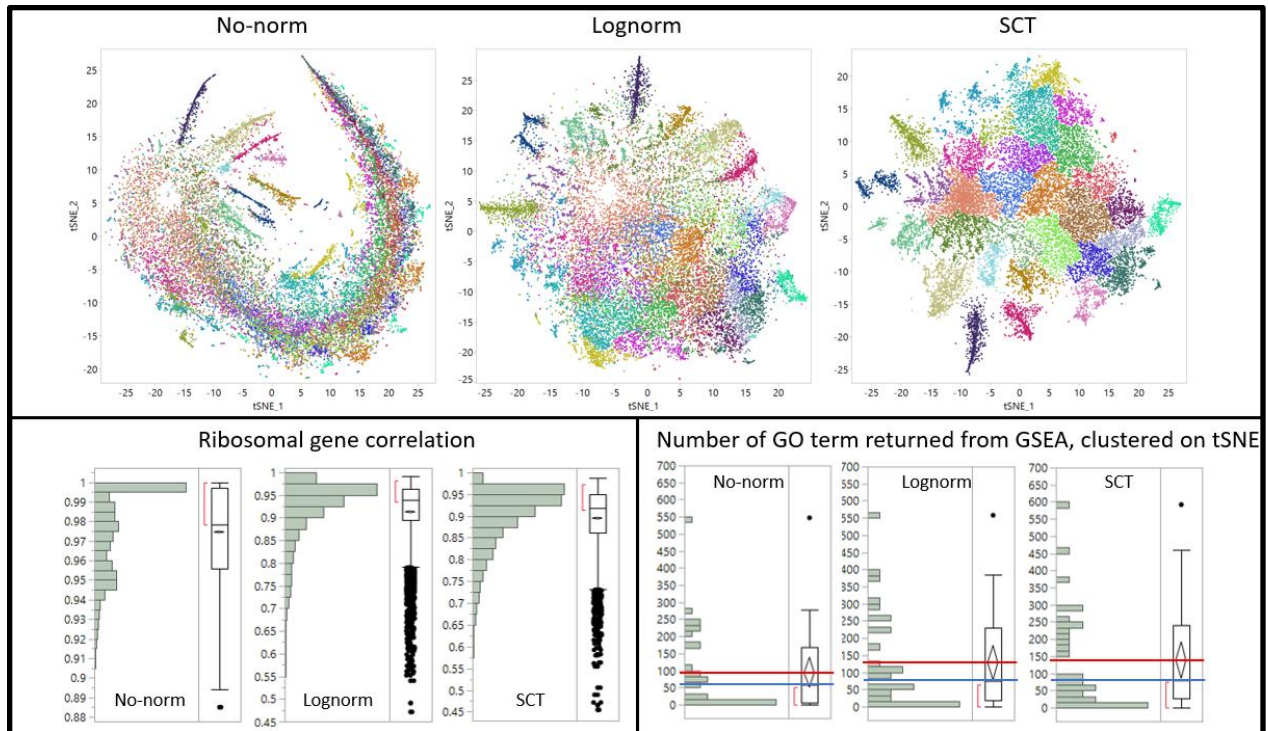
The ribosomal protein gene set is ubiquitously expressed in all cells at high level, gene-gene correlation within this set could be used to assess library quality (REF?). The other two sets of genes is mutually exclusive in single cell (REF and FIG), if metagene properly extracted, the gene-gene correlation within each set should also be high.

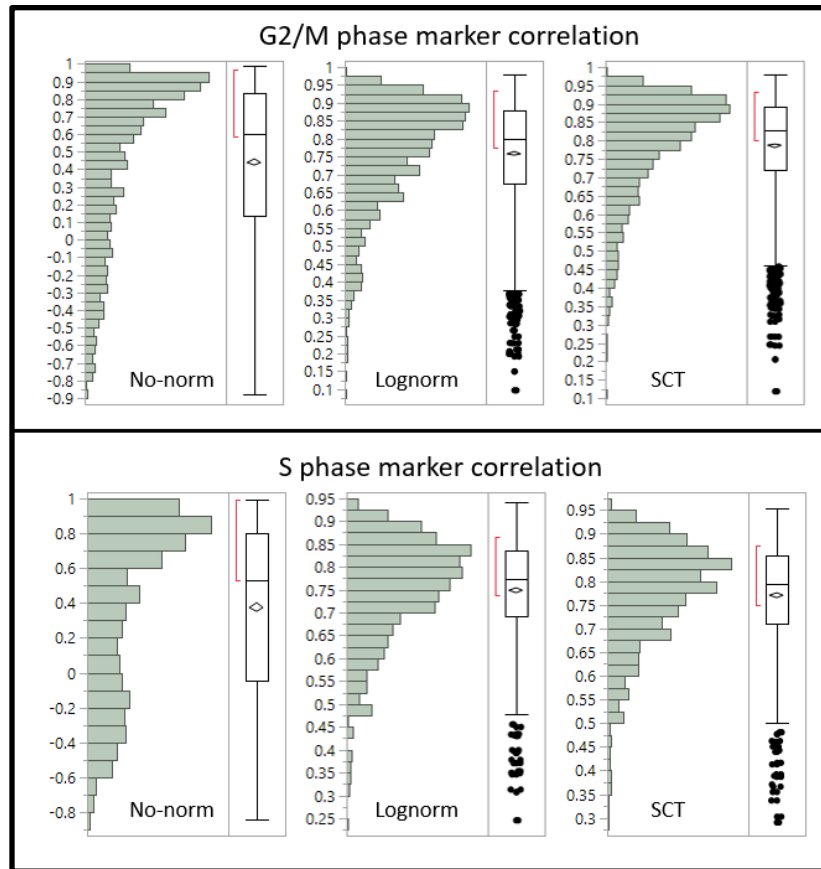
Key modifications on workflow is described below:

1.1. Normalization data for MG extraction

Since sequencing depth varies between libraries, normalization is one important step in scRNA seq pipeline to remove such bias (REF). To test if normalization could improve the quality of metagene, we compared results from dataset processed in 3 different ways: non-normalized, log-normalized (REF), SCT-normalized (REF). TSNE projection of all three sets were made after PCA, and colored by Ward clustering of SCT-normalized dataset (FIG). Based on colored tSNE projection, the structure of some isolated clusters are quite concordant across all three sets, while the body part of non-normalized dataset shows very different structure compared to the other two. For gene-gene correlation, the ribosomal protein genes show high correlation in all three sets, but cell cycle genes correlation of non-normalized dataset is significantly lower than normalized datasets [FIG]. Then metagenes are extracted from each dataset from corresponding tSNE and

evaluated by GSEA. Again non-normalized dataset shows both low mean and median number of GO:BP terms returned. As conclusion, normalization is necessary to remove bias from different libraries for metagene extraction. Using SCT compared to canonical log normalization only improve marginally, but since it provides better theoretical foundation and tighter clusters, we used SCT normalization in our workflow.

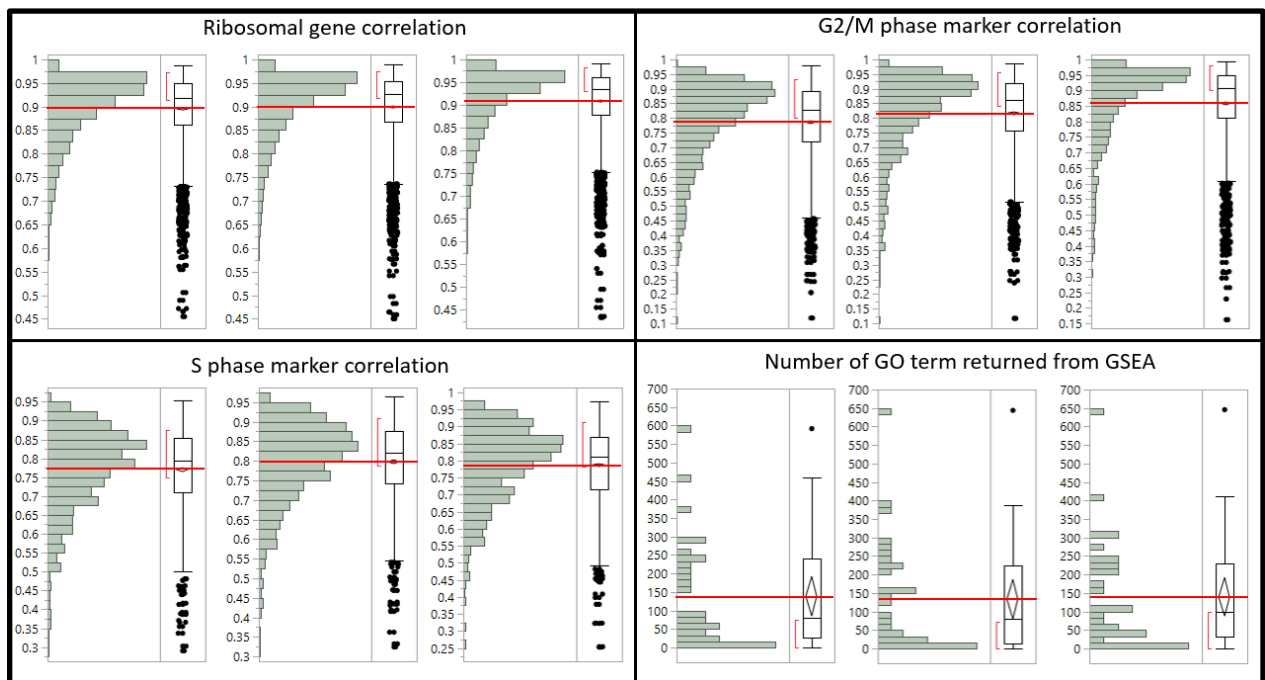
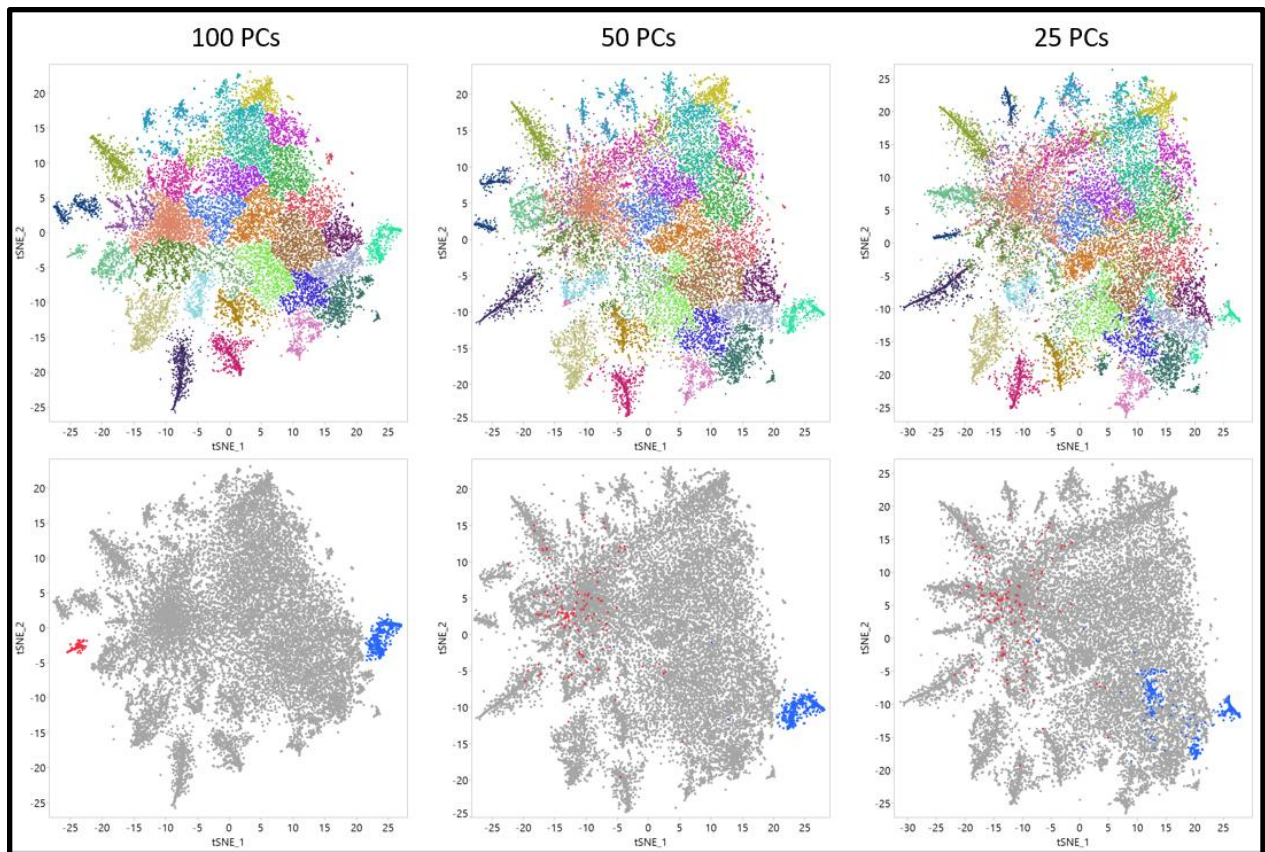




1.2. Optimization of the number of principal components used

In canonical scRNA seq pipeline, various approaches have been used to determine the number of PCs used for downstream processing, of which the most widely used heuristic approach is ranking the percent variance explained by each PC. Usually the number of PCs is dataset dependent, and should have explained a large percentage of variance. To optimize the number of PCs used in our metagene extraction workflow, we compared tSNE structures from different number of PCs. By projecting Ward Hierarchical clustering assignment from 100 PC resulted tSNE to other tSNE space shows no significant difference and vice versa [Fig]. However, when we dive into each individual cluster, we find examples that only identified when more PCs used. Further GSEA on these cluster shows the genes have related GO:BP terms enriched, indicating that these genes are

functional related. For example, one cluster enriched in [] related genes can only be identified when 100 PC used. Another cluster enriched in cell cycle related genes also does not isolate if only 25 PCs used. Although determining the dimensionality of a dataset is challenging and subjective, we conclude that compared to canonical scRNA seq pipeline, more PCs are needed to precisely capture functional related genes.



1.3. Clustering of genes is different from clustering of cells

Once the dimensionality of the input dataset is properly reduced and visualized in tSNE space, the next challenge is to put genes into meaningful groups by clustering. Clustering approaches for cells have been introduced in the previous chapter [\[LINK\]](#), but a few modifications are needed for genes.

First, the reduced dimension space of genes is heavily biased by gene expression and detection level [\[Fig\]](#), genes with low expression/detection forms a cloud in left center while genes with high expression detection forms another continuous cloud on the other side. As functional related genes don't necessarily have the same expression/detection level, clustering in PC space is also biased [\[Fig\]](#).

Second, the structure of genes in tSNE space is much more complicated than that of cells. Giving that functional gene groups could be varied in size, canonical Ward Hierarchical clustering might not work well.

Instead of clustering in reduced dimensions (PC or tSNE), we choose to compute pairwise correlation of genes from PCs and cluster on correlation. The isolated clusters are nicely kept, and by evaluating GSEA results, more enriched GO terms are captured compared to clustering in PC or tSNE space [\[Fig\]](#). Other than that, we used a modified Ward Hierarchical clustering approach named Iterative Binary Splitting (IBS) to perform clustering [\[Fig\]](#). In this approach, the dataset is split by Ward Hierarchical clustering into two clusters, each resulting cluster is then split again by Ward Hierarchical clustering, iteratively. By doing so we will get 2^N clusters, N being the number of iterations. There are several advantages of IBS compared to Ward:

1. No pre-assumption of cluster number needed.
2. Each split is an independent event and will not be affected by other clusters.

3. Flexible cluster size and stopping point. Splitting of one branch could be stopped while other branches keep splitting.

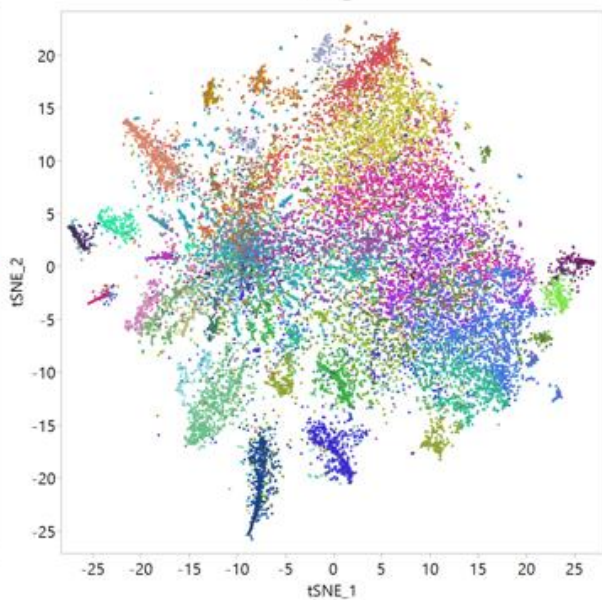
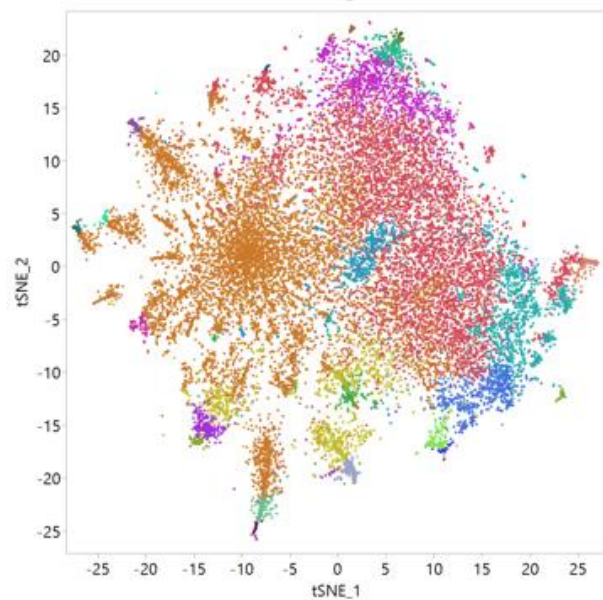
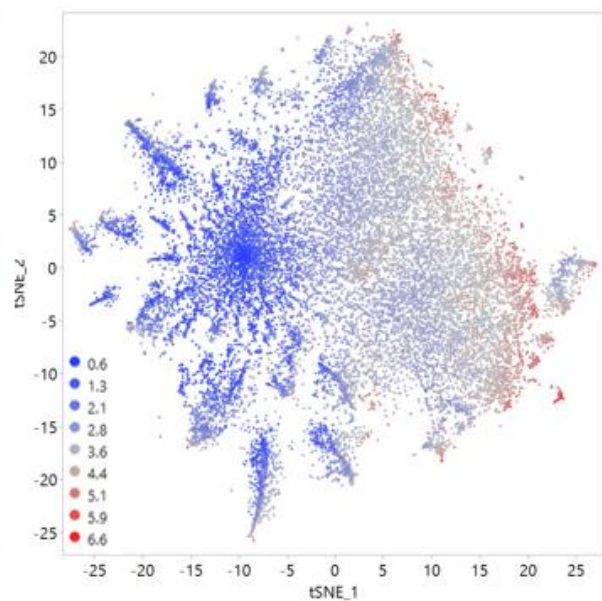
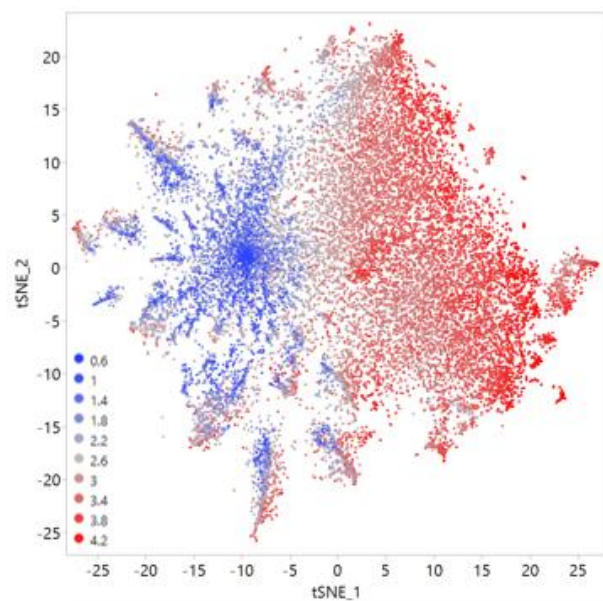
4. Other advantages.

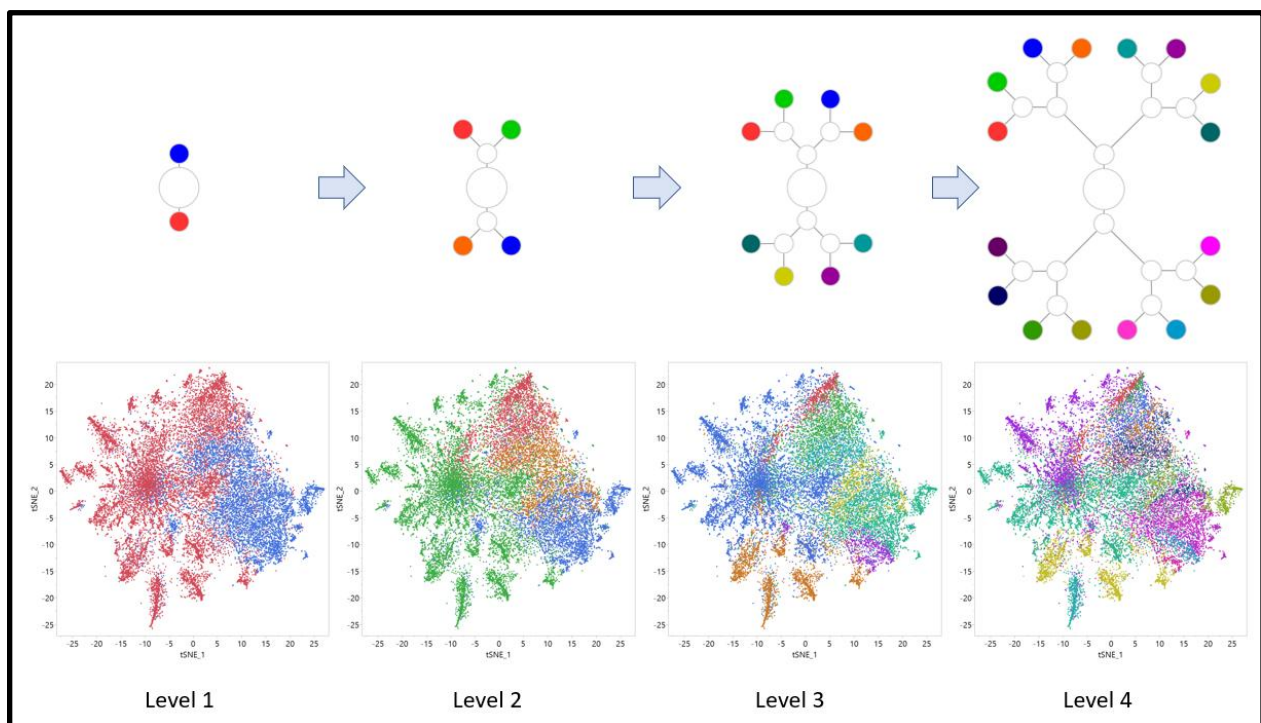
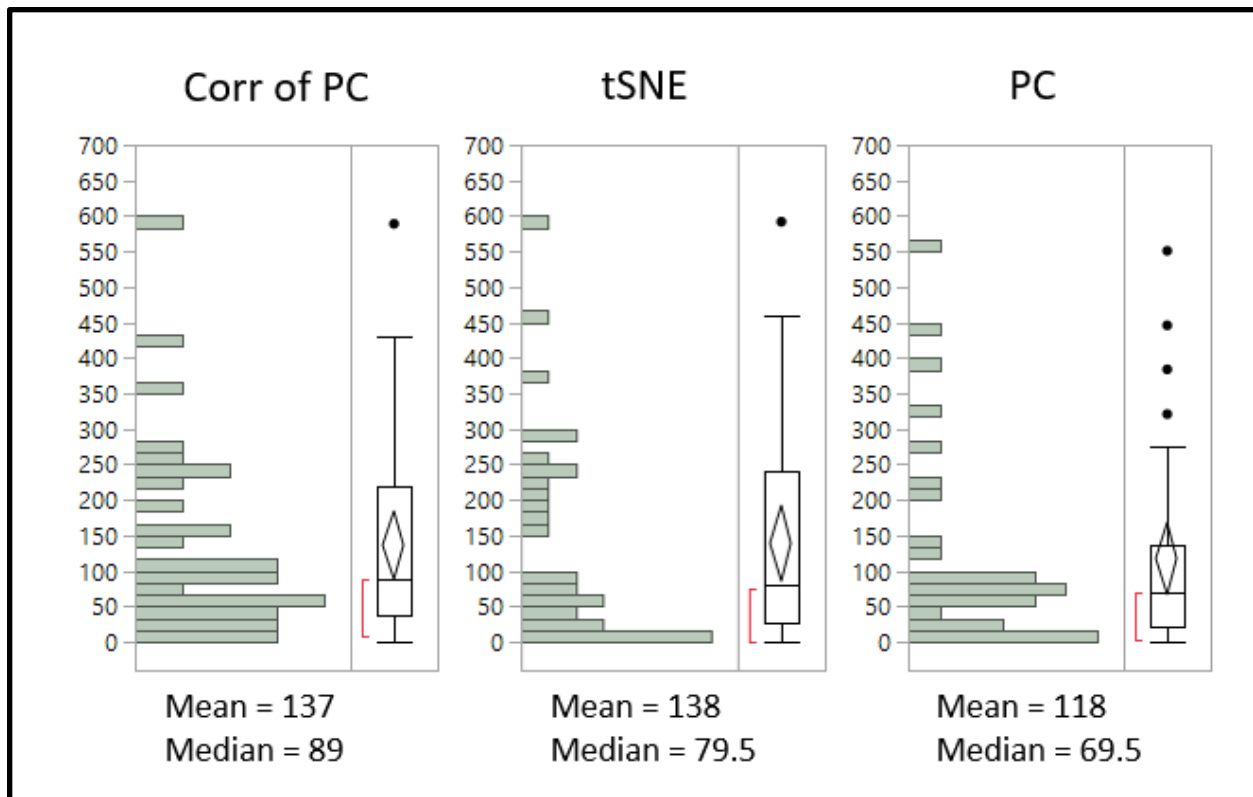
Here we applied IBS to our dataset up to 10 iterations (1024) clusters. The clustering result is then trimmed by manually assigning stopping points to each split, based on the following general rules:

1. If two daughter clusters show similar GOBP enrichment results, they should be stopped at mother cluster.
2. If both daughter clusters show significant lower GOBP enrichment p-values, they should be stopped at mother cluster.

3. [Other criteria we use in determining stopping point]

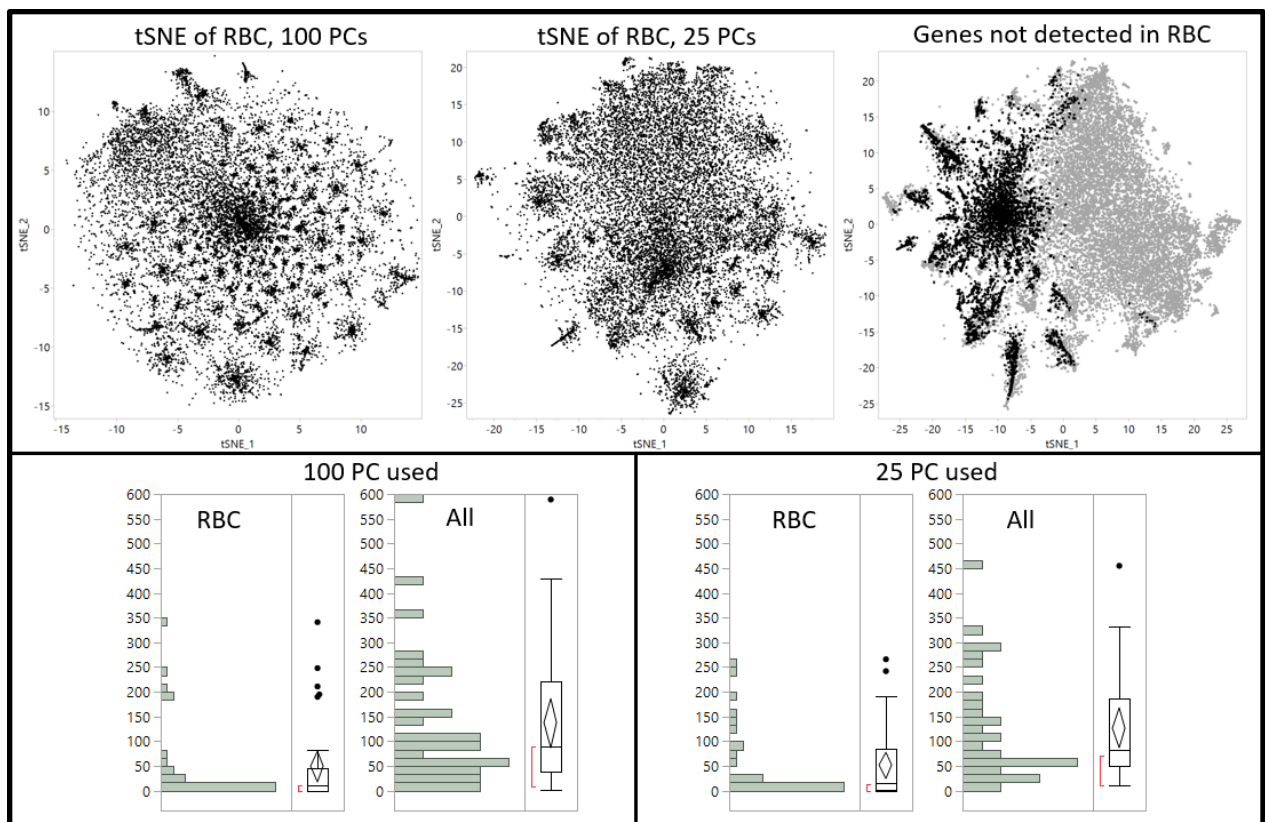
We finalized with 63 gene clusters, which shows improved GOBP enrichment results compared to Ward [Fig].

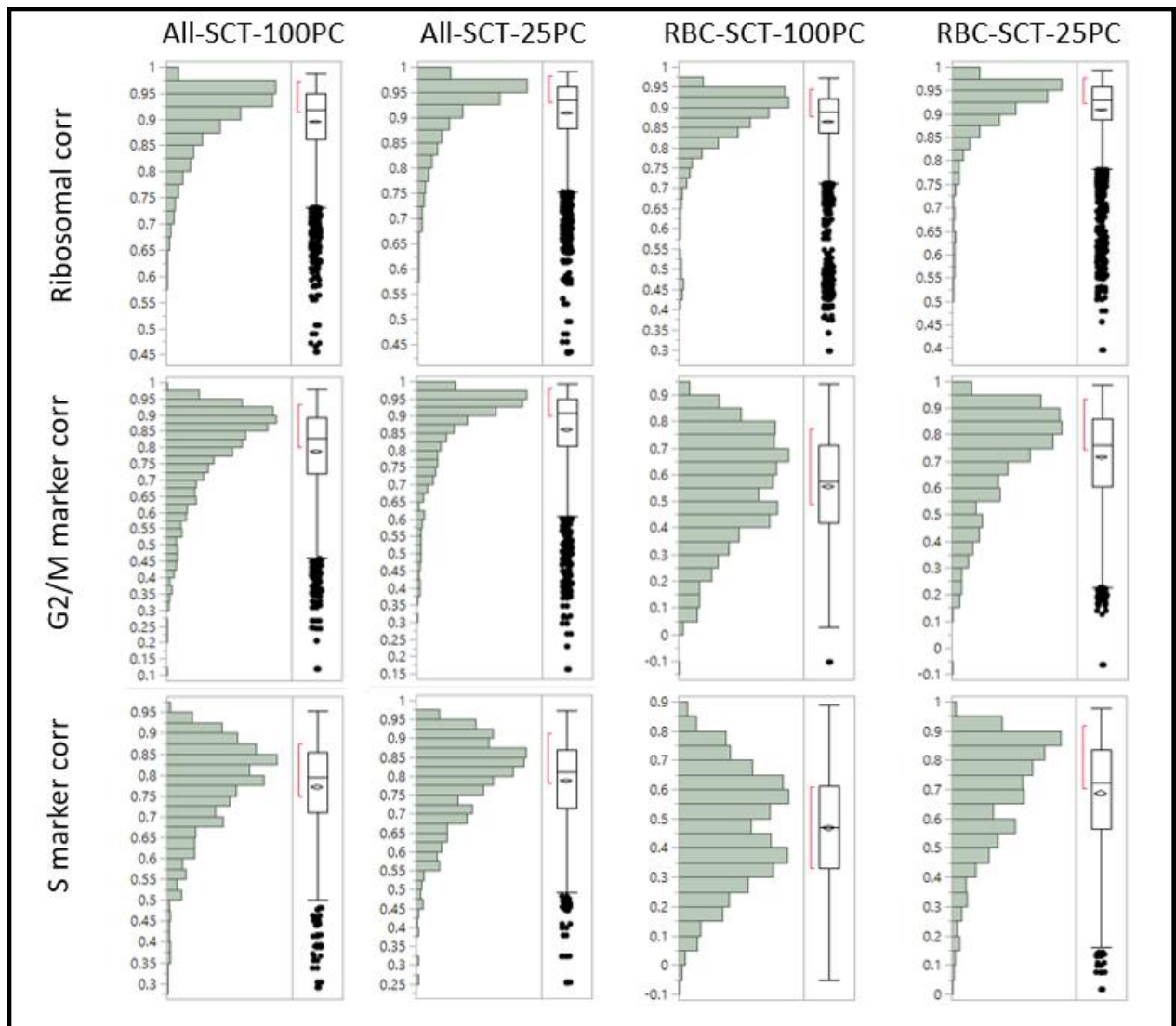




1.4. Highly heterogeneous dataset is needed for metagene extraction

Unlike pipeline designed for cell clusters, the pipeline for metagene extraction requires input dataset with high heterogeneity. It's easy to understand as certain genes only express in specific cell types, thus cannot be identified if there's no such cell in dataset. To test this, we performed metagene extraction on a dataset with only red blood cells, compared to our TS19 whole embryo dataset. Considering reduced variance in less heterogeneous dataset, both 25 and 100 PCs used. Compared to whole embryo dataset, the tSNE projection from RBC-only dataset shows less structures. Specifically, the cluster-like structures seen in the RBC projection are not real meaningful functional-related genes based on GSEA results, which come from tSNE artifact when inappropriate dimensions selected [Fig]. Lots of genes were dropped because of low expression/detection in RBC [Fig], and further analysis of GSEA results from RBC dataset does not identify gene clusters with meaningful functions other than only a few housekeeping gene clusters.





In summary, our metagene extraction workflow can be summarized as below:

- I. Highly heterogeneous dataset obtained from scRNA seq.
- II. Data normalization by SCT transformation and transposition.
- III. Dimension reduction by PCA, then tSNE from top 100 PCs.
- IV. Clustering from gene-gene correlation of 100 PCs using IBS.
- V. Biological interpretations by GSEA.

2. Cell type specific genes and housekeeping genes identified from TS19 mouse embryo metagenes

Using workflow described above, we extracted 63 metagenes from TS19 dataset. Except for 9 clusters with weak or mixed signals assigned for 'undefined', GSEA results for all metagenes shows there are two big categories of metagenes: 26 cell type specific metagene and 28 ubiquitously expressed metagene (or housekeeping metagenes). Comparing GSEA results with gene detection/expression plot shows good match: housekeeping metagenes are expressed in various cell clusters while cell type specific metagenes shows cell cluster specific expression pattern [Fig]. In tSNE space, the cell type specific metagenes are mostly represented by well isolated clusters, while housekeeping metagenes are connected except two specific clusters. Cell type specific genes show unique patterns in PC space as their expression level in certain cell type differs significantly depending on cell, thus easier to be isolated. Most housekeeping genes can be detected in almost all cells with one exception as cell cycle genes, which could also be isolated from housekeeping metagene cloud [Fig]. Another exception is ribosomal genes, as their expression level is so high that reaches almost 100% detection rate, they are tightly correlated and clustered [Fig].

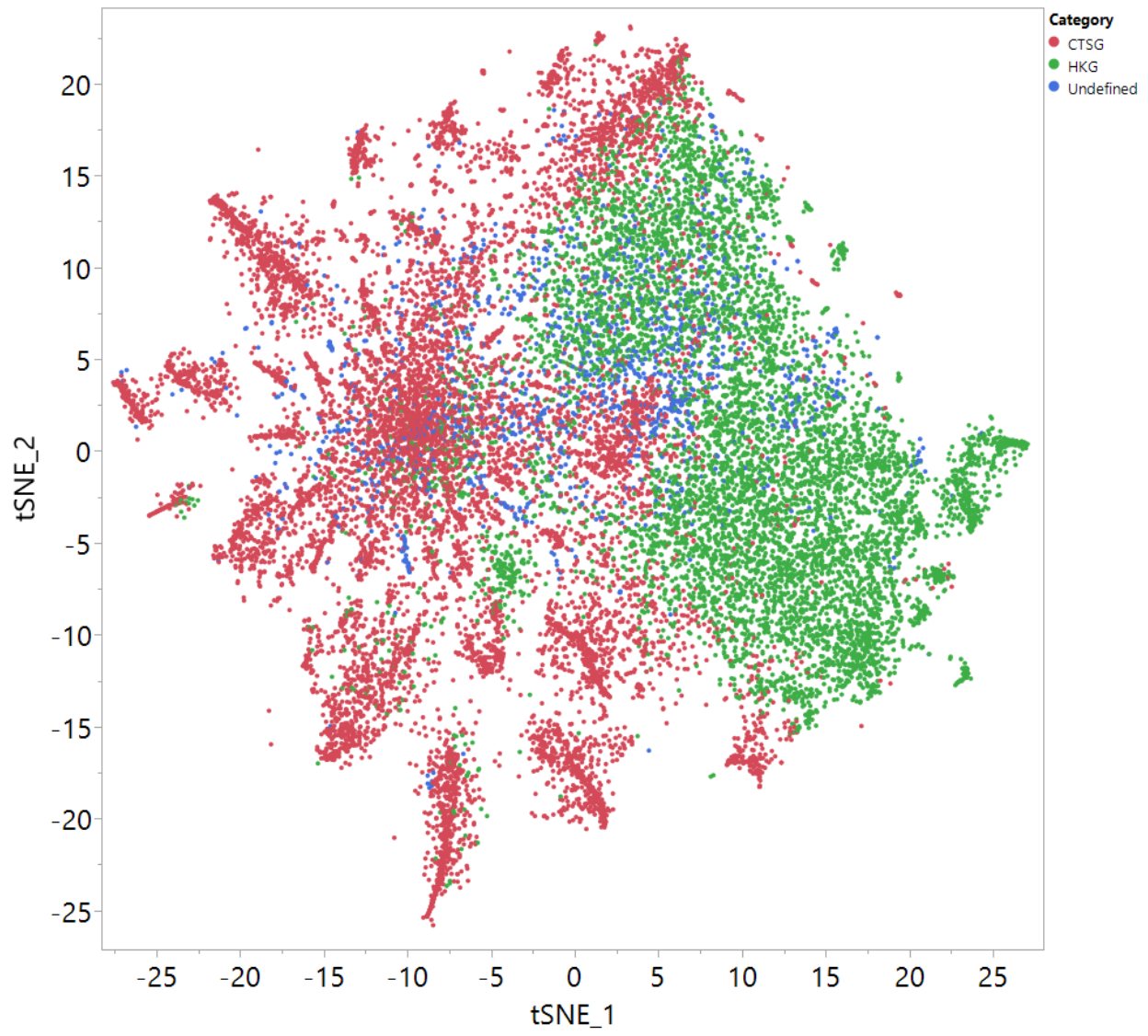
To better understand metagene structure and visualization, we created an IBS tree, showing 63 terminal nodes after trimming [Fig]. To understand each metagene, we project the genes into gene tSNE space, also we made cell tSNE projection using canonical scRNA seq pipeline for cell and project gene expression level of each metagene to cell tSNE space. As expected, most cell type specific metagene highlight specific cells in cell

tSNE space, while housekeeping metagenes highlight most of the cells. Projections and GSEA interpretation of each metagene is provided in supplemental figures in detail.

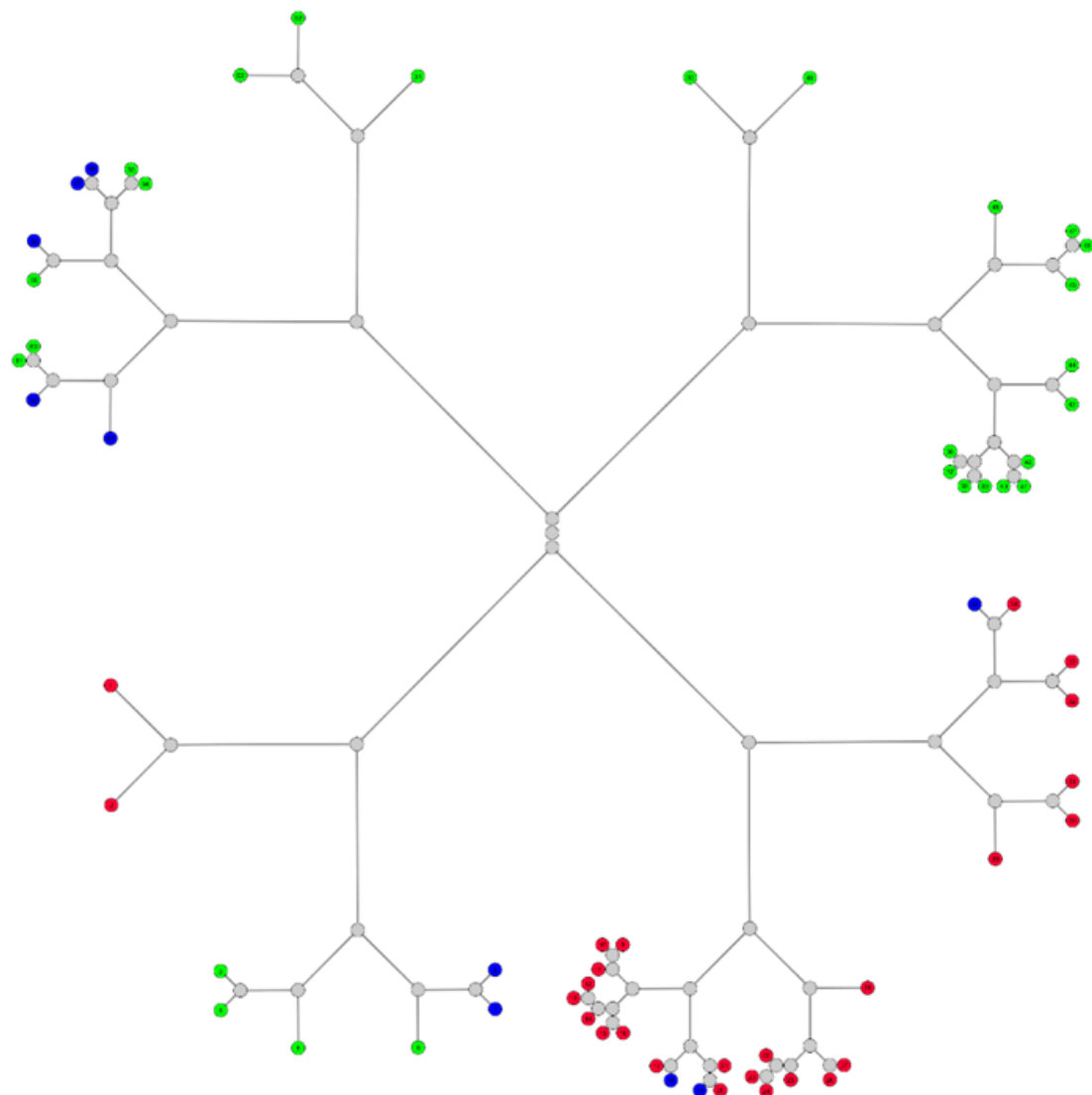
For 26 cell type specific metagenes we extracted, we have [REDACTED].

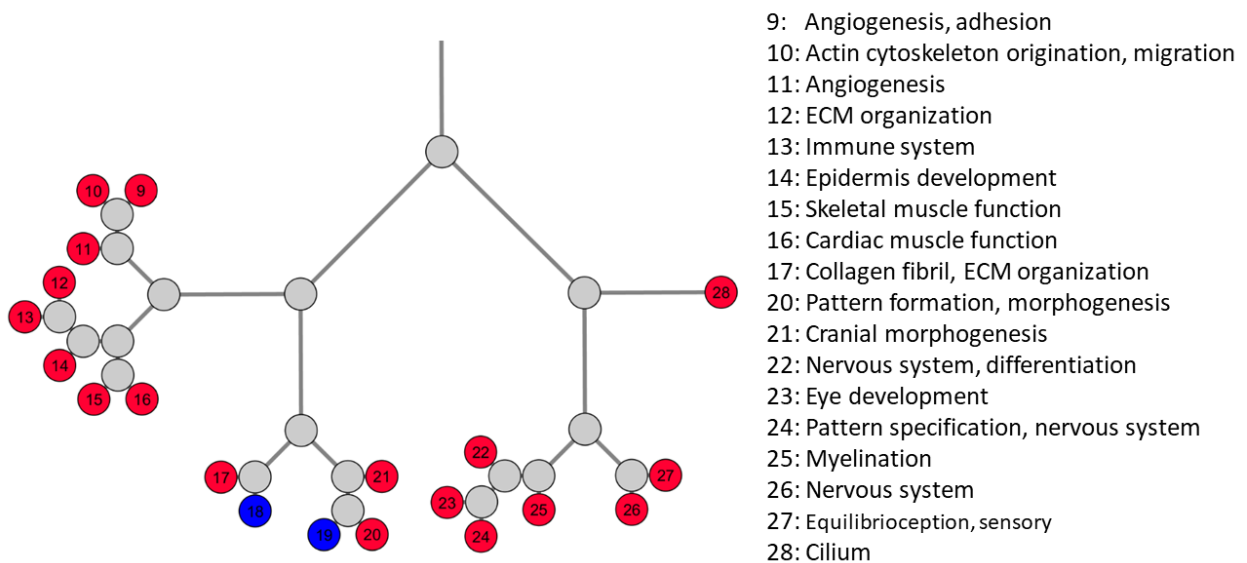
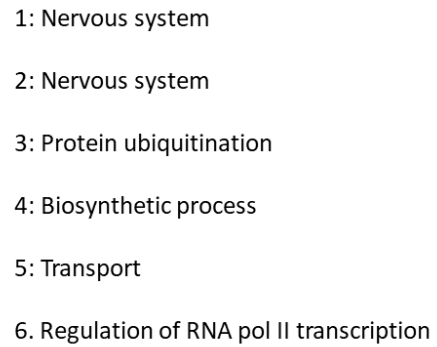
For 28 housekeeping metagenes we extracted, we have [REDACTED].

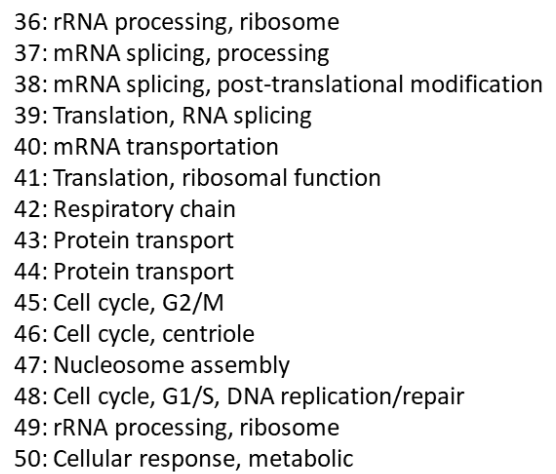
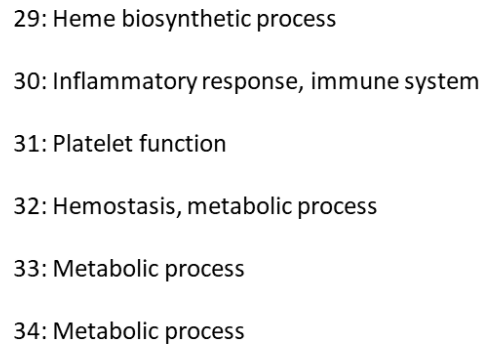
By analyzing all metagene GSEA results and projections, we confirmed that our pipeline can precisely extract metagene from scRNA seq data. One extreme evidence is a cluster of histone genes, both detection and expression level of most histone genes are very low due to the nature limitation of scRNA seq technology, on average only 1% of the cells detect these genes, and only 1 transcript per detected cell. But through our pipeline, we identified a metagene with Z% histone genes, which shows high sensitivity.

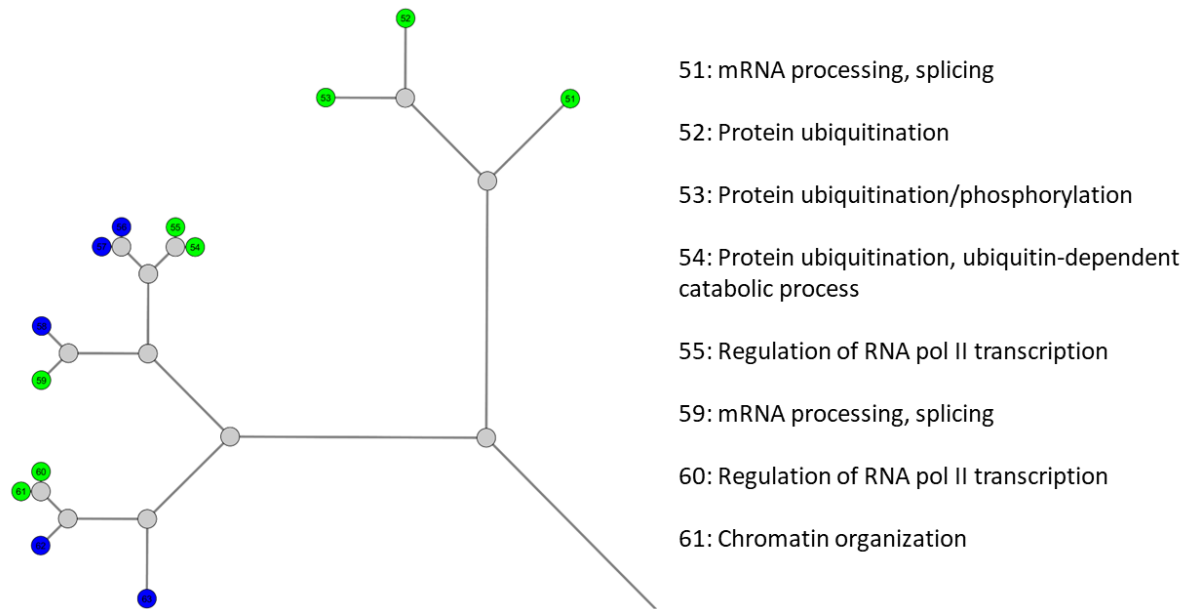


(need modification on undefined)









3. Use metagene to understand cell cycle regulation

Of all the metagenes identified, two of which show strong signal of cell cycle related GOBP terms. More specifically, one metagene is related to G1/S phase of cell cycle and the other is related to G2/M. Projecting specific cell cycle gene marker from S phase and G2/M to tSNE space show high concordance with metagenes [Fig]. Correlation between markers and other genes within metagene also indicate that these two metagenes are involved in S phase and G2/M phase of the cell cycle.

By calculating the expression of two marker gene lists at individual cell level, we confirmed that these genes are mutually exclusively expressed [Experiment and figure].

We further investigate all genes within the metagene by using GeneCards. For metagene cluster 45 annotated as G2/M phase metagene, we identified 194 genes, 49 of which are in G2/M marker gene list. For metagene cluster 48 annotated as S phase metagene, we identified 237 genes, 41 of which are in G2/M marker gene list. To prioritize the

understudied genes in metagene, we computed various matrices listed below to help rank genes: [redacted]

Specifically, we computed pairwise correlation between each gene and two cell cycle marker lists individually. By ranking the average correlation, we can tell which genes are potentially highly co-expressed with known cell cycle markers. We manually analyzed top 150 genes correlated to S phase markers by GO terms and literature descriptions, 125 genes show functions directly related to S phase, 11 genes show some clues but no direct evidence related to S phase, 14 genes show almost no available data or unrelated functions. For top 150 genes correlated to G2/M phase markers, 124 genes show functions directly related to G2/M phase, 9 genes show some clues but no direct evidence related to S phase, 12 genes show almost no available data or unrelated functions, and 5 genes turned out to be non protein coding genes.

4. Identification of genes involved in mouse epicardium development

Of all the metagenes identified, one of which shows very specific expression level in cell space while GSEA returns poor p-values [Fig]. This might indicate this metagene is cell type specific, but many members are not well annotated by GO. Based on limited information from GOBP, we hypothesize that the metagene is probably related to epicardium develop, with several epicardium marker genes found this metagene. [Gene XYZ from REF considered as markers found in metagene, also gene ABC involved in mouse heart development found in metagene...] [Fig]. By running GSEA of corresponding cell cluster highlighted by this metagene, we also see evidence showing these cells are

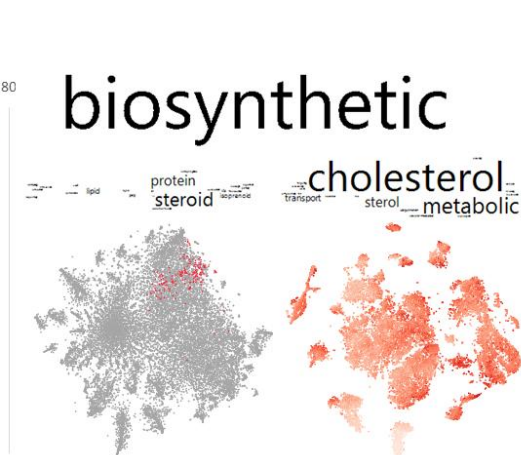
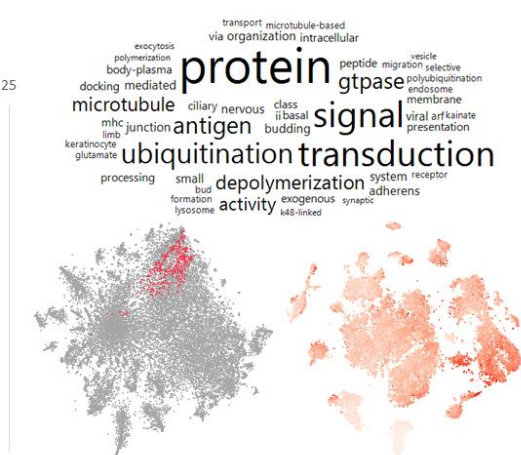
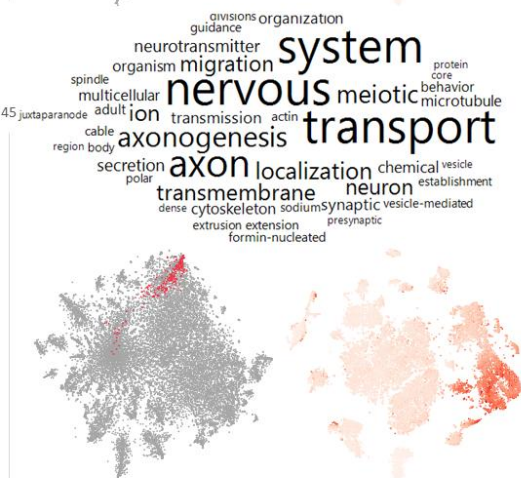
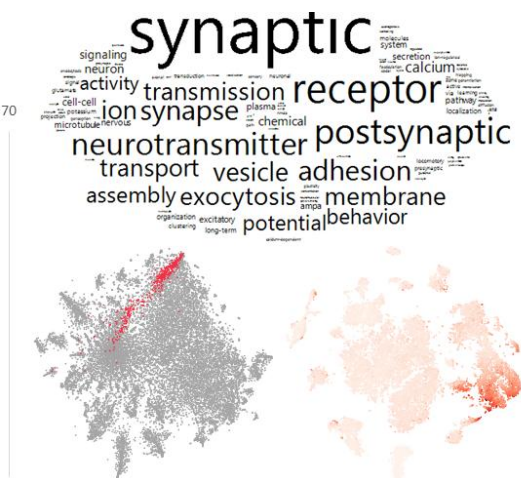
epicardium cells [Fig]. We prioritized the genes within metagene using previously described approach and tested gene [redacted] via in-situ hybridization.

5. Genes controlling cell type differentiation observed in metagene

Of all the metagenes identified, we see two metagenes related to cardiac muscle and skeletal muscle development. To investigate the relationship between these two metagenes, we isolate these two clusters and further analyze members from these metagenes.

Discussion

- Cluster-like structure observed in MG projection of RBC



Biological Process	$-\log(p\text{-value})$
Transferrin Transport	24.5
Receptor Signaling Pathway	24.0
Increased Oxygen Levels	21.0
Induction of Macroautophagy	19.5
Endosome-mediated Transport	17.5
Plasma Membrane Transport	17.0
Coupled Proton Transport	16.5
Presentation of Exogenous Antigen by MHC Class II	15.5
Macroautophagy	13.5
Endosome-mediated Transport	11.5

Figure 1 displays two network diagrams illustrating the enrichment of biological processes. The left diagram (grey) shows a dense, interconnected network of terms, primarily related to protein synthesis and transport, with 'protein' and 'transferrin' as central nodes. The right diagram (orange) shows a more dispersed network of terms, primarily related to cellular signaling and metabolism, with 'cellular signaling' and 'metabolic' as central nodes. Both diagrams include a scale bar indicating the number of genes (1000).

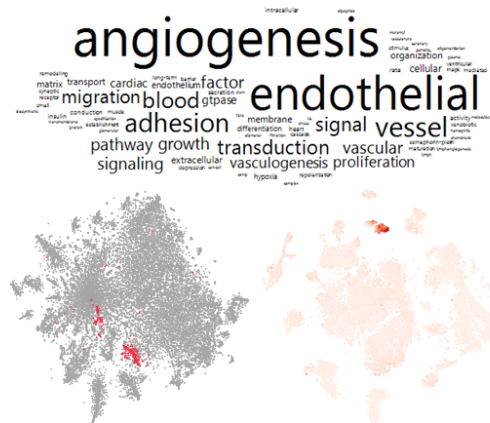
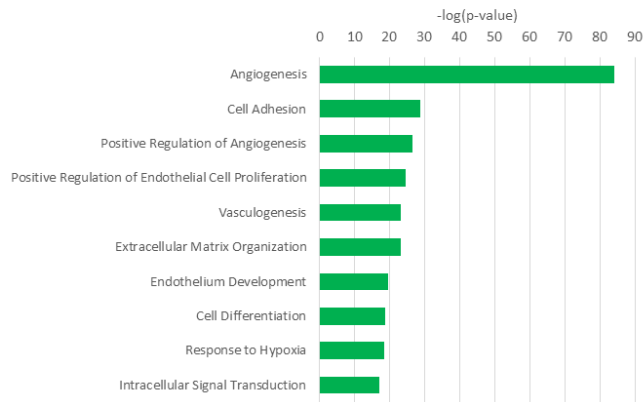
Biological Process	$-\log(p\text{-value})$
Transcription By RNA Polymerase II	~30
Chromatin Organization	~23
Transcription, DNA-templated	~16
Biological_process	~14
Histone H3-K27 Methylation	~13
Brain Development	~12
Cilium Assembly	~11
Gene Expression, Epigenetic	~10
Transcription, DNA-templated	~10
Chromosome Condensation	~10

Biological Process	$-\log(p\text{-value})$
Via Plasma Membrane	~28
Cell Adhesion	~20
Synaptic Transmission	~19
Synapse Assembly	~19
Adhesion Via Plasma Membrane	~16
Methylation Via Threonine	~12
Signaling G Protein-coupled	~11
Signaling Pathway	~11
Transmission, GABAergic	~10
Signaling G Protein-coupled	~10
Signaling Pathway	~10
Dendrite Self-avoidance	~10

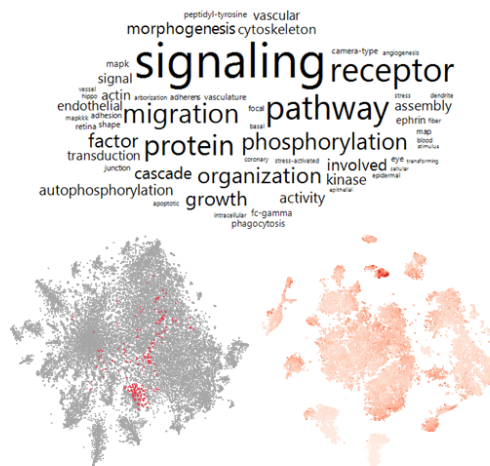
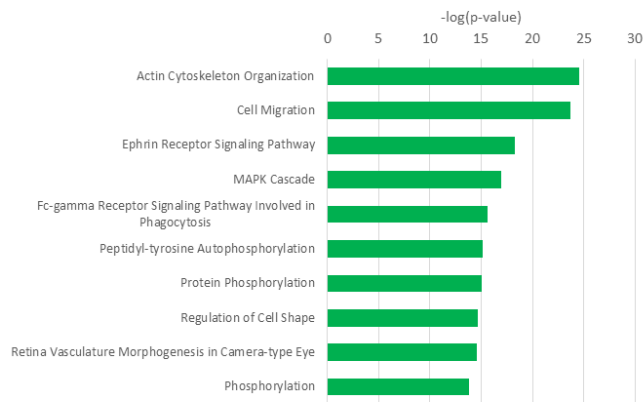
Biological Process	$-\log(p\text{-value})$
Autophagy of Mitochondrion	~15.5
Protein Phosphorylation	~10.5
Response to Nitrogen Starvation	~9.5
Glycogen Metabolic Process	~9.5
Regulation of Cilium Assembly	~9.0
Mitochondrion Organization	~8.5
Long-term Synaptic Potentiation	~8.5
Extension of Branchiomotor Axon	~7.5
Transmembrane Transport	~8.0
Cysteine Transport	~8.0

glucocorticoid assembly transport nitrogen
metabolic phosphorylation cd24 cellular mammary
microvillus organization starvation transmembrane
mediated division prolactin
receptor somatic

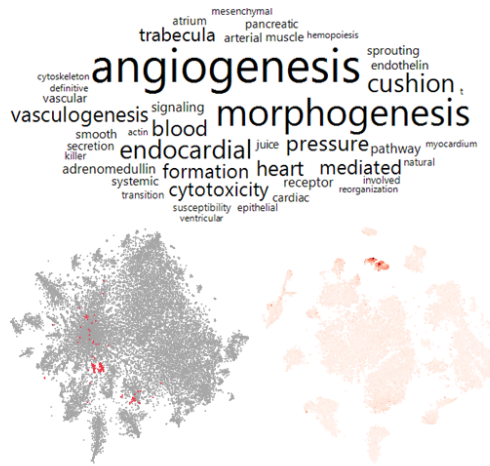
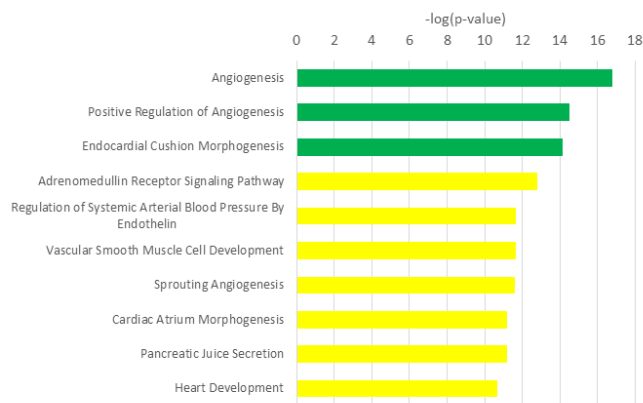
GO:BP enrichment from cluster 9



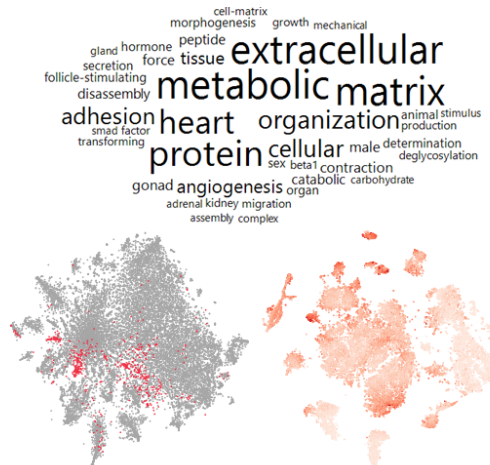
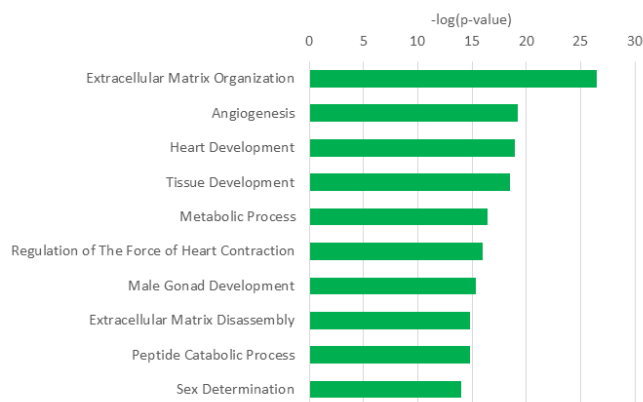
GO:BP enrichment from cluster 10



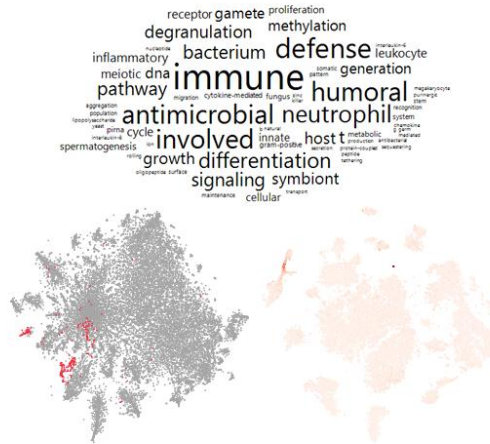
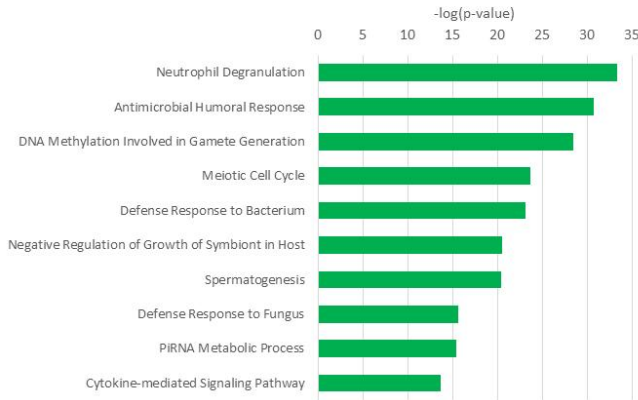
GO:BP enrichment from cluster 11



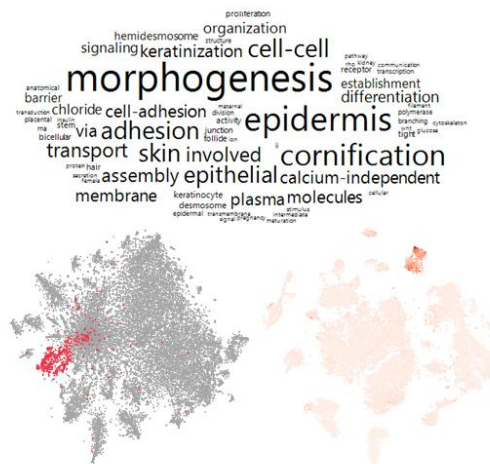
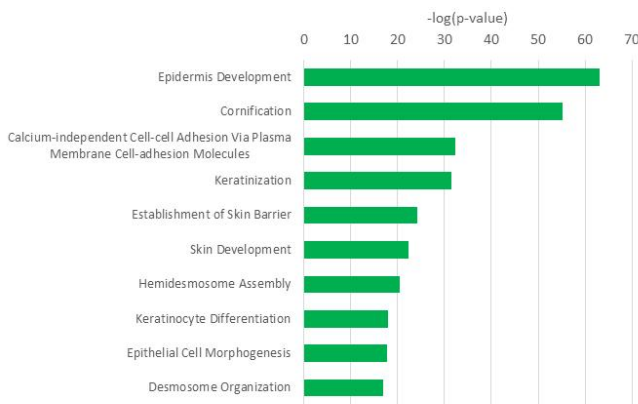
GO:BP enrichment from cluster 12



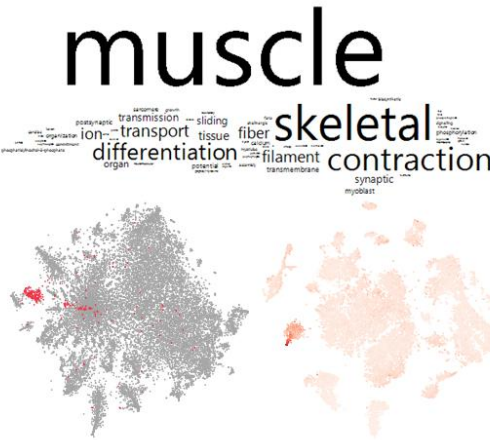
GO:BP enrichment from cluster 13



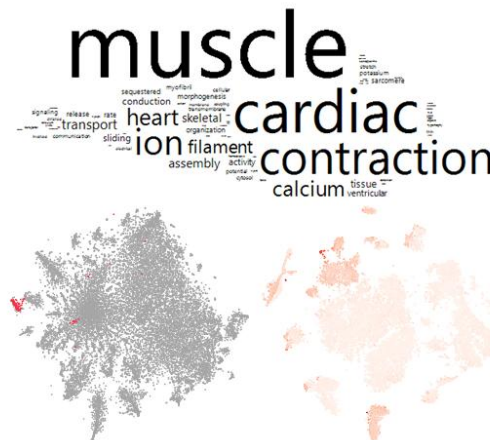
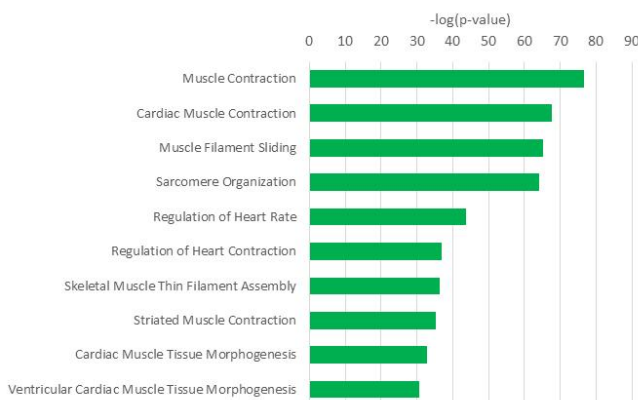
GO:BP enrichment from cluster 14



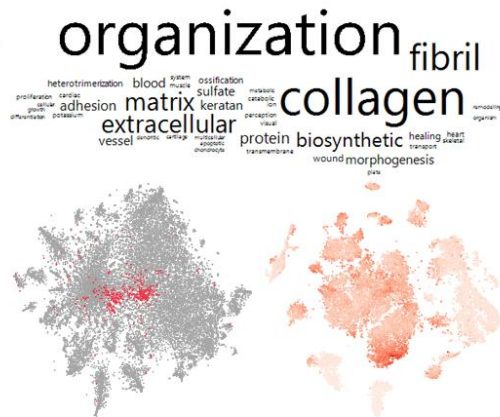
GO:BP enrichment from cluster 15



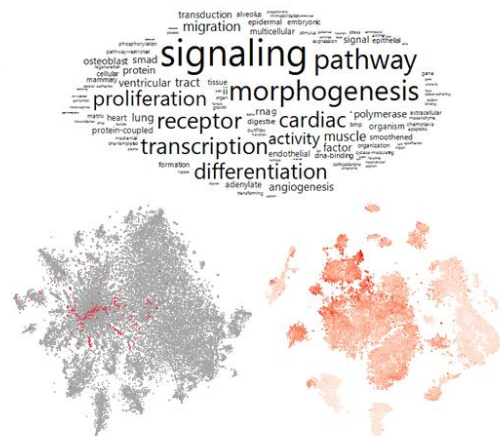
GO:BP enrichment from cluster 16



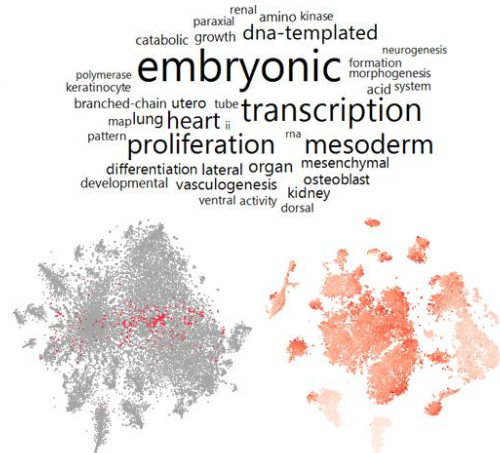
Biological Process	$-\log(p\text{-value})$
Collagen Fibril Organization	~65
Extracellular Matrix Organization	~45
Cell Adhesion	~25
Collagen Biosynthetic Process	~23
Wound Healing	~18
Protein Heterotrimerization	~17
Ossification	~16
Heart Morphogenesis	~14
Keratan Sulfate Catabolic Process	~13
Blood Vessel Development	~13



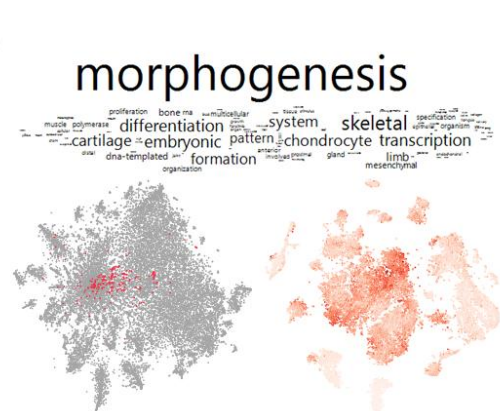
GO Term	$-\log(p\text{-value})$
Regulation of Signaling Receptor Activity	~30.5
Multicellular Organism Development	~28.5
Lung Development	~26.5
Positive Regulation of Cell Proliferation	~23.5
Negative Regulation of DNA-binding Transcription Factor Activity	~22.5
Extracellular Matrix Organization	~21.5
Positive Regulation of Transcription By RNA Polymerase II	~21.0
BMP Signaling Pathway	~20.5
Positive Regulation of Cardiac Muscle Cell Proliferation	~20.0
Positive Chemotaxis	~18.0



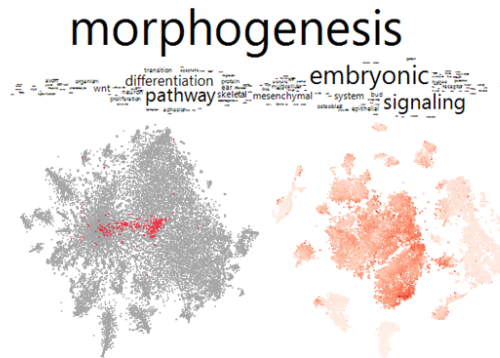
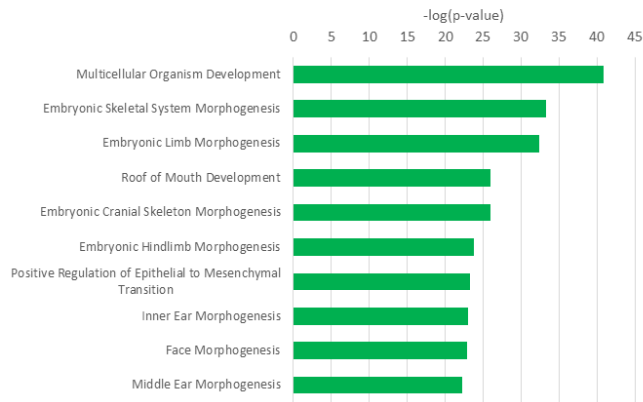
GO Term	$-\log(p\text{-value})$
Positive Regulation of Transcription, DNA-templated	~19.5
Embryonic Organ Development	~16.5
Lung Development	~16.0
Lateral Mesoderm Development	~15.5
Vasculogenesis	~15.0
Positive Regulation of Mesenchymal Cell Proliferation	~14.5
in Utero Embryonic Development	~14.5
Regulation of Osteoblast Differentiation	~14.0
Kidney Development	~14.0
Developmental Growth	~13.5



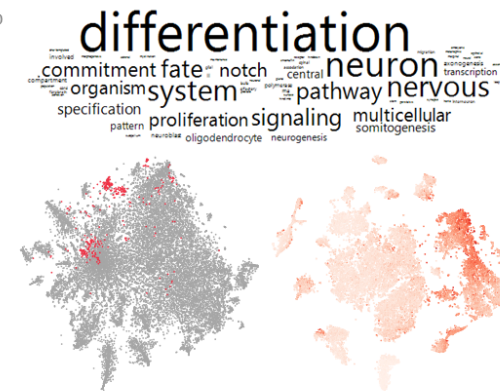
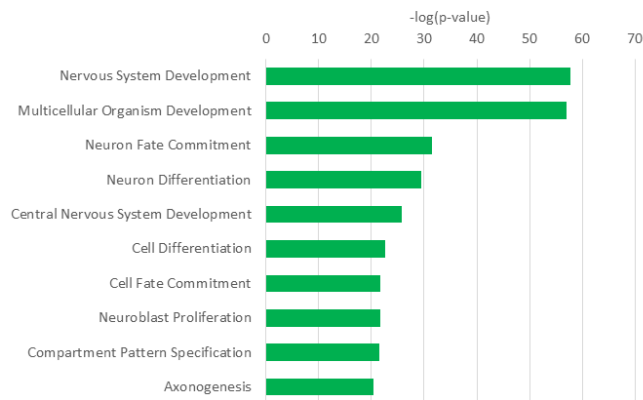
Biological Process	$-\log(p\text{-value})$
Skeletal System Development	~70
Multicellular Organism Development	~45
Proximal/distal Pattern Formation	~35
Embryonic Limb Morphogenesis	~35
Regulation of Transcription, DNA-templated	~33
Chondrocyte Differentiation	~31
Positive Regulation of Transcription By RNA Polymerase II	~29
Cartilage Development	~28
Positive Regulation of Chondrocyte Differentiation	~27
Embryonic Skeletal Joint Morphogenesis	~26



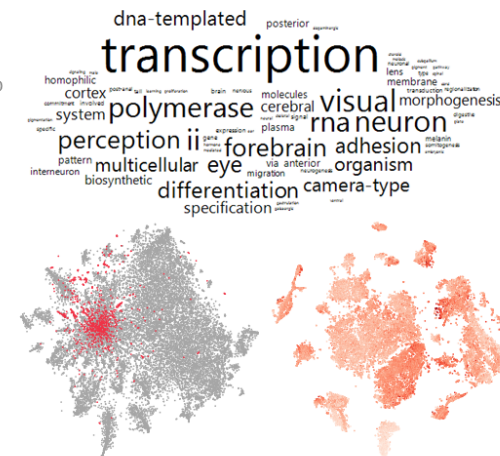
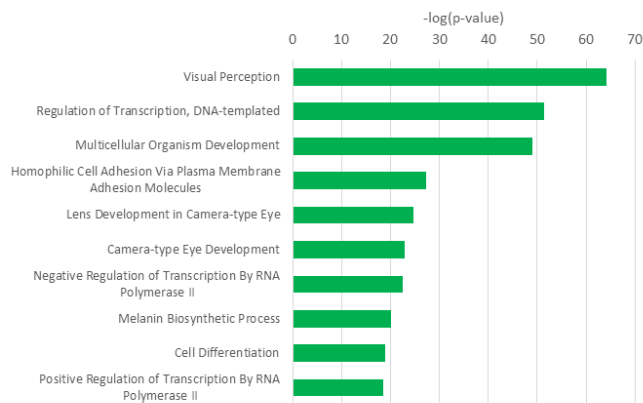
GO:BP enrichment from cluster 21



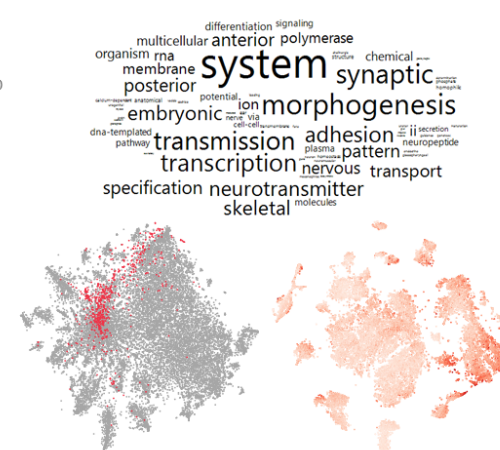
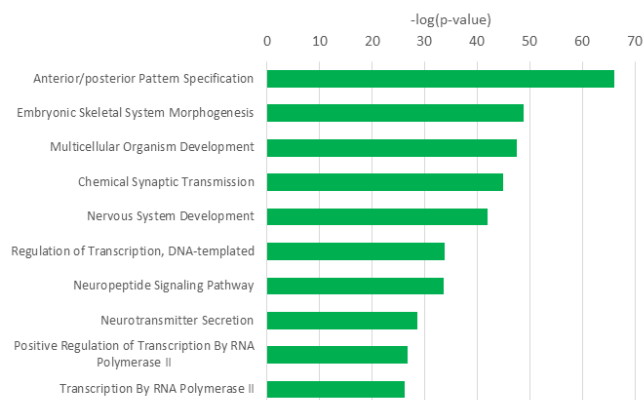
GO:BP enrichment from cluster 22



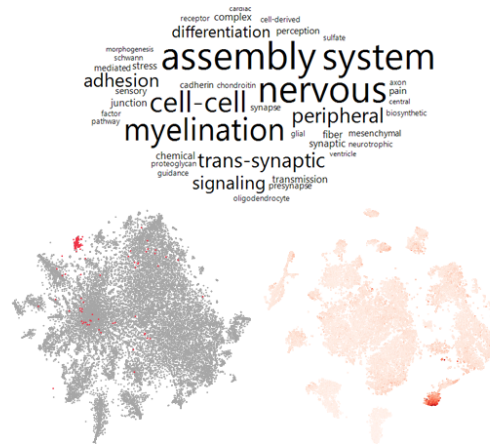
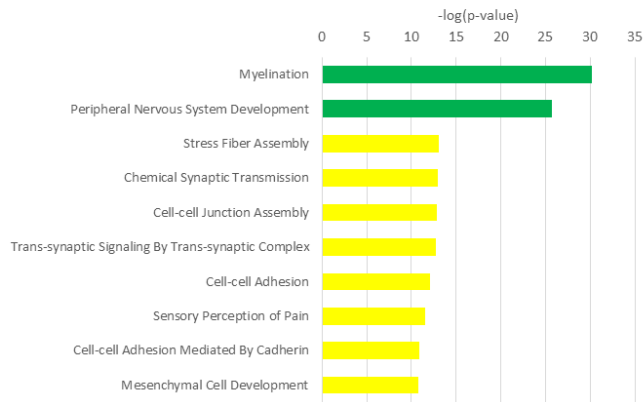
GO:BP enrichment from cluster 23



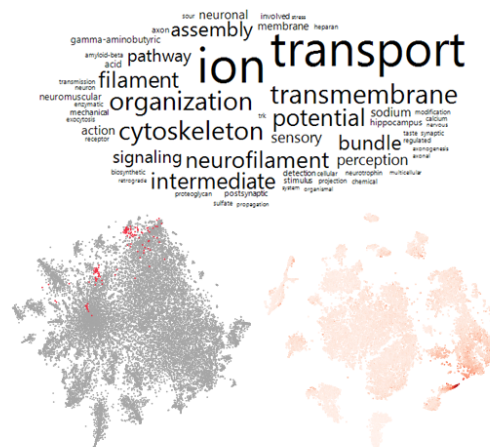
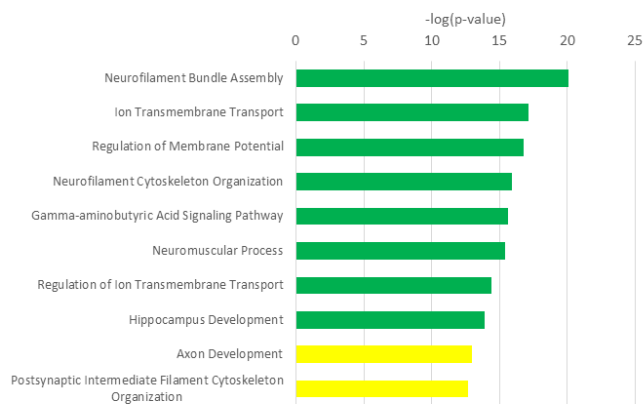
GO:BP enrichment from cluster 24



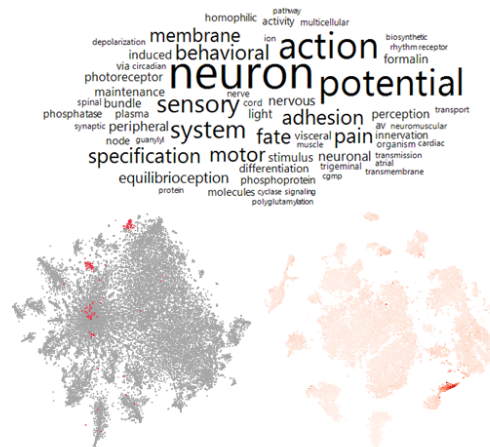
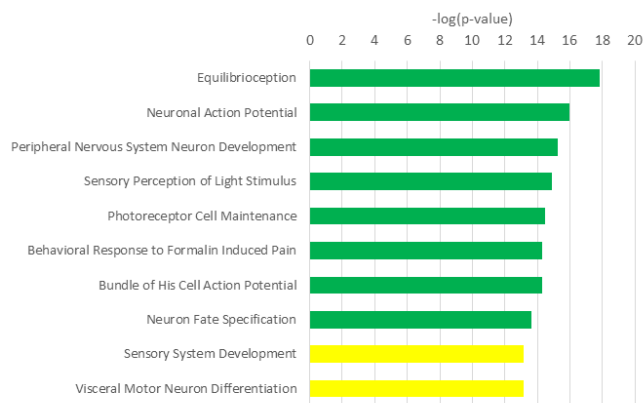
GO:BP enrichment from cluster 25



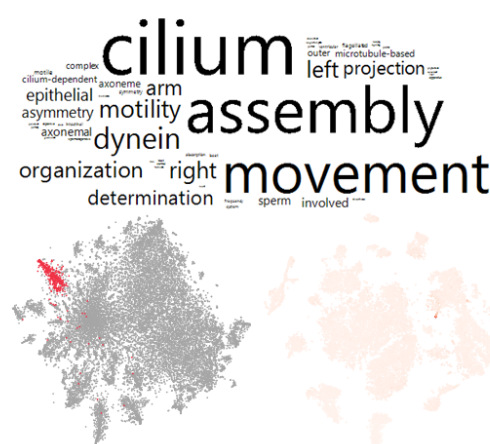
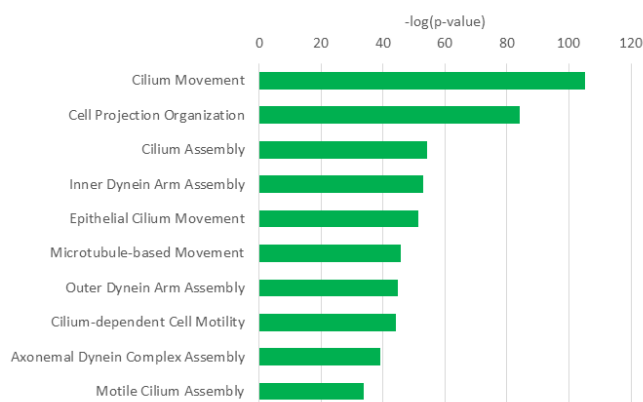
GO:BP enrichment from cluster 26



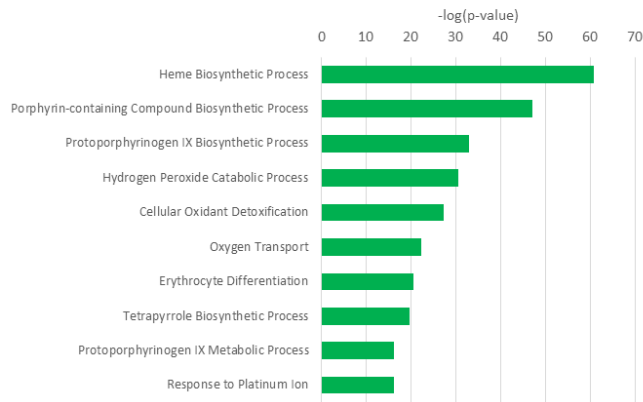
GO:BP enrichment from cluster 27



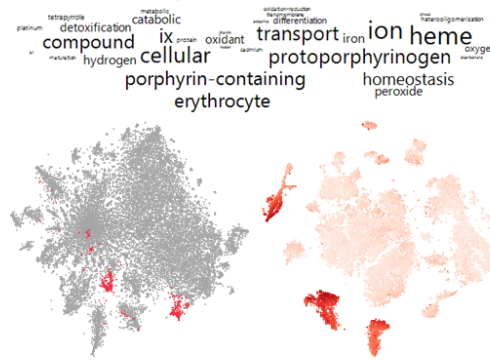
GO:BP enrichment from cluster 28



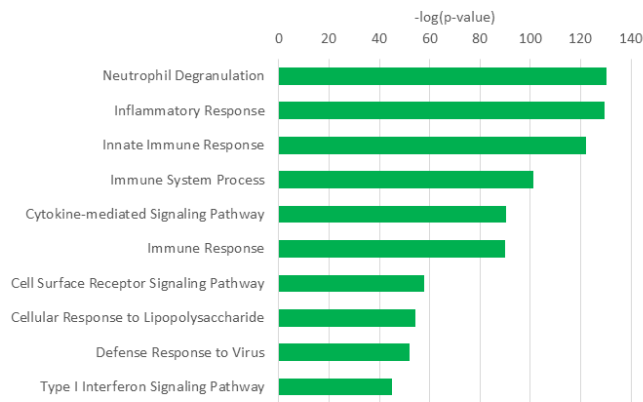
GO:BP enrichment from cluster 29



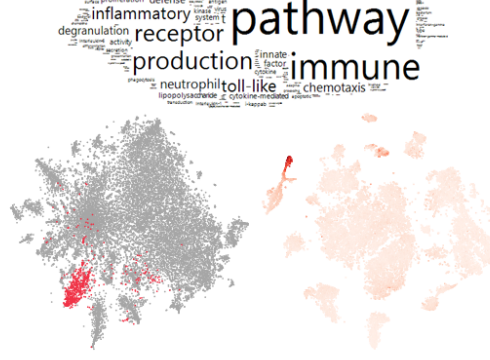
biosynthetic



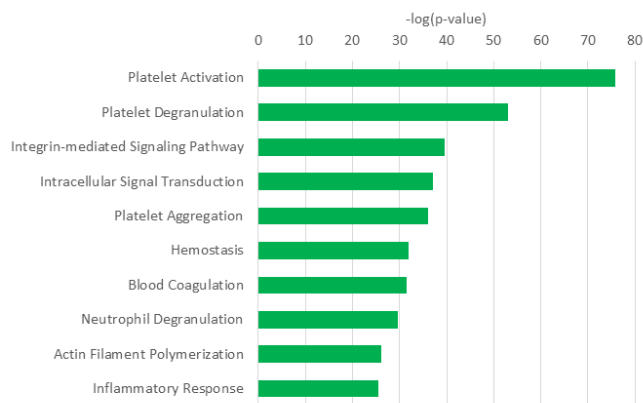
GO:BP enrichment from cluster 30



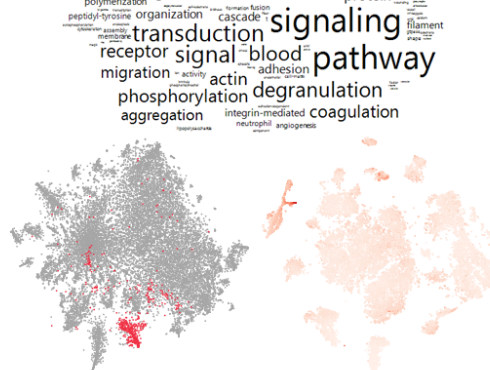
signaling



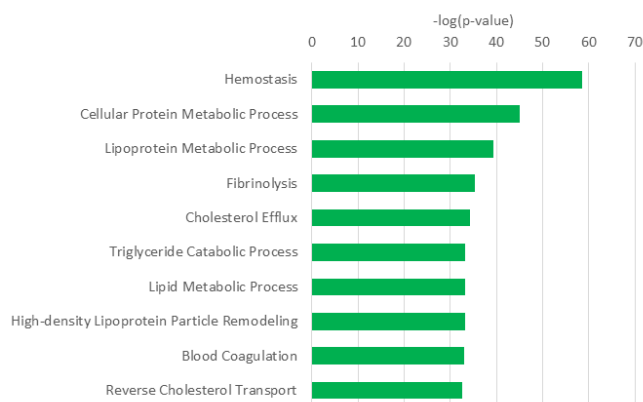
GO:BP enrichment from cluster 31



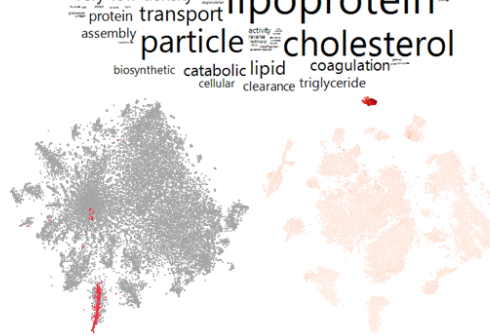
platelet



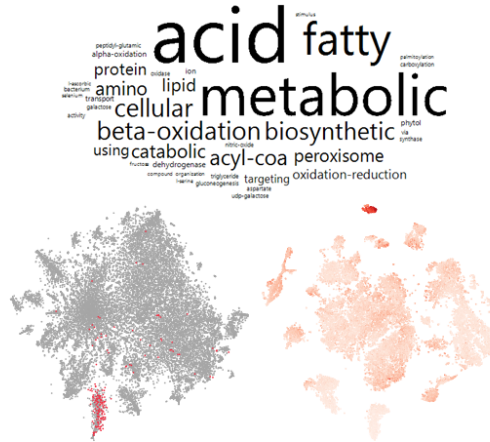
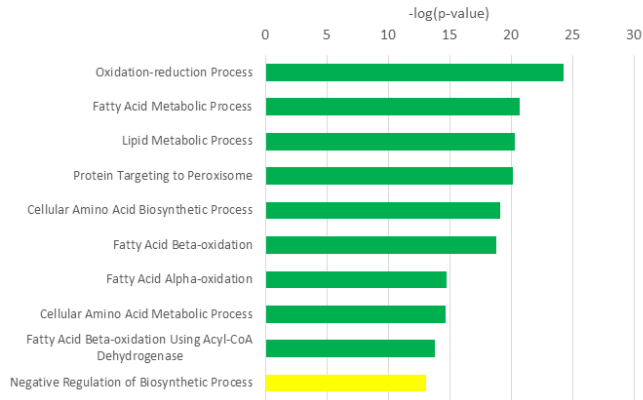
GO:BP enrichment from cluster 32



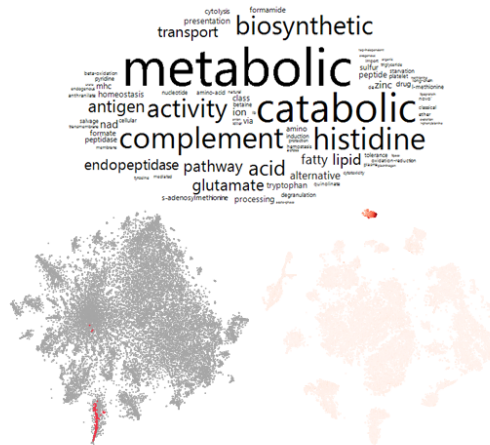
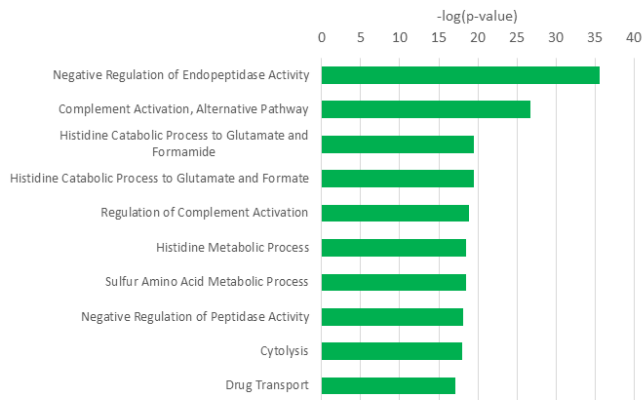
metabolic



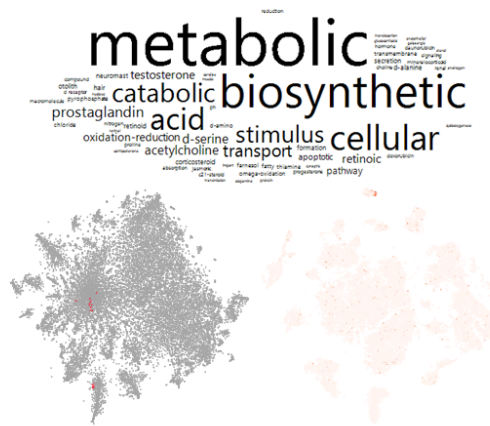
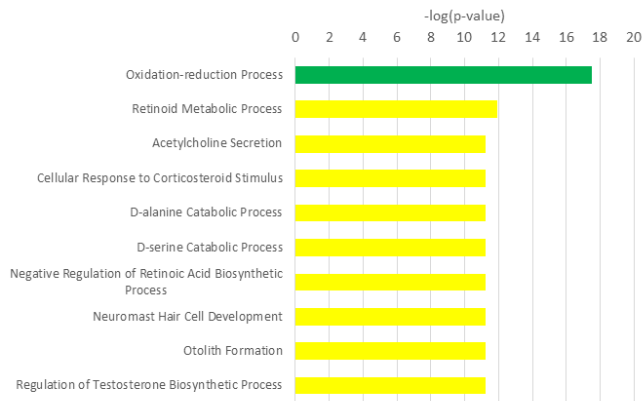
GO:BP enrichment from cluster 33



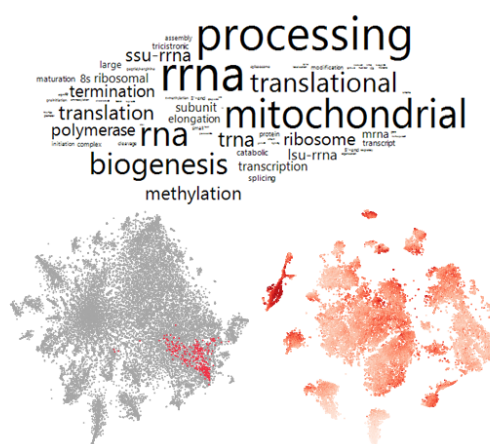
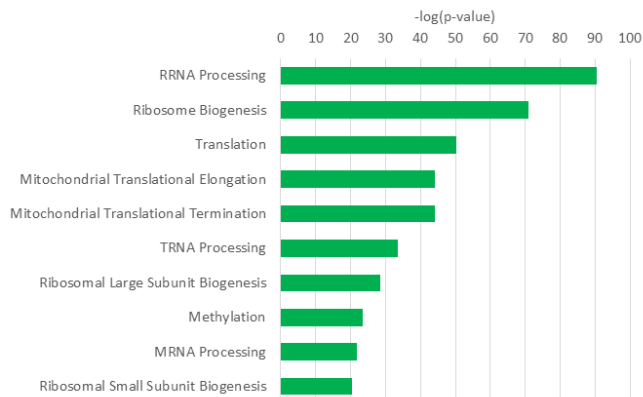
GO:BP enrichment from cluster 34



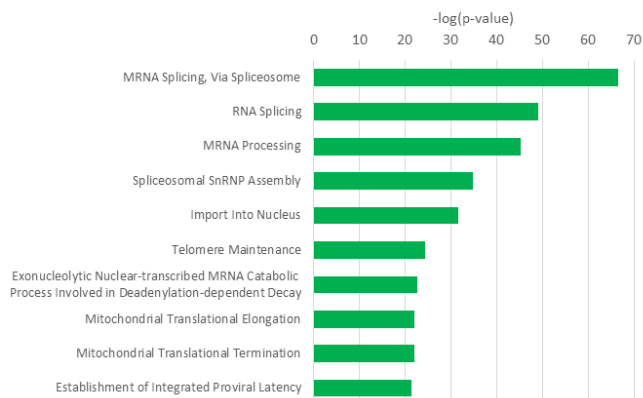
GO:BP enrichment from cluster 35



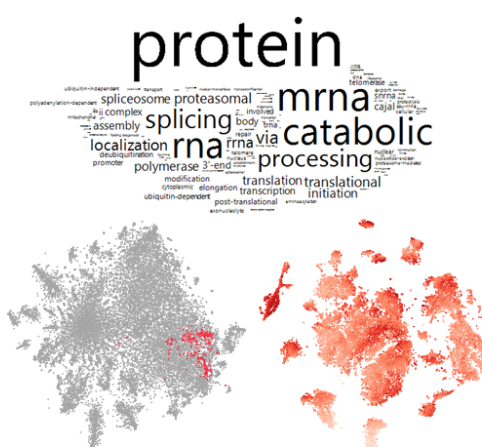
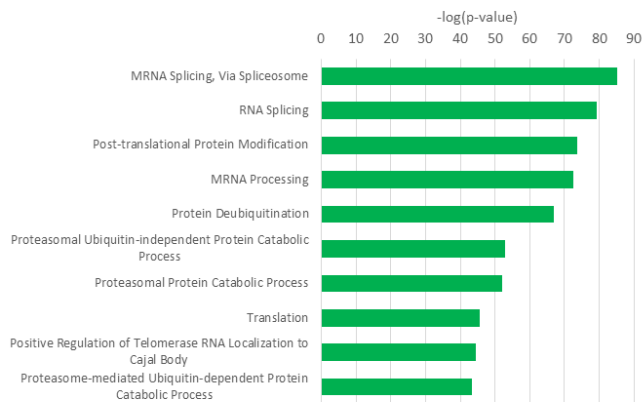
GO:BP enrichment from cluster 36



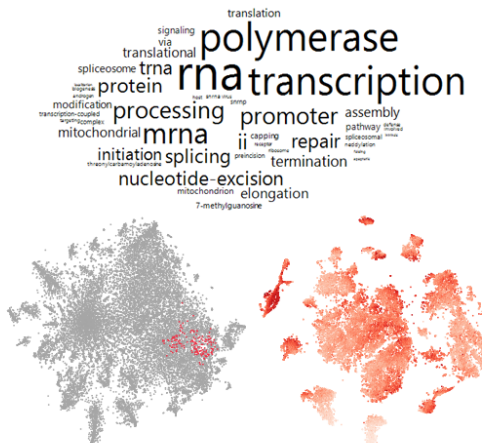
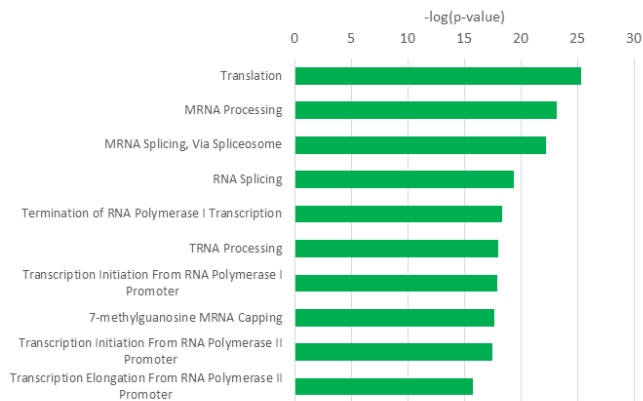
GO:BP enrichment from cluster 37



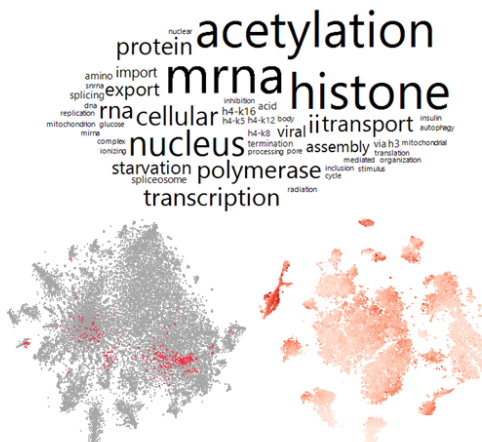
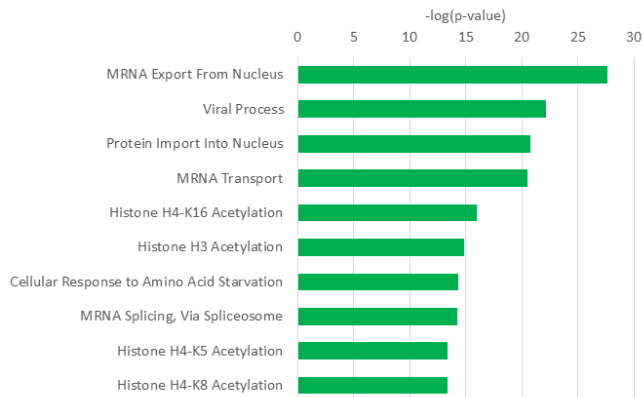
GO:BP enrichment from cluster 38



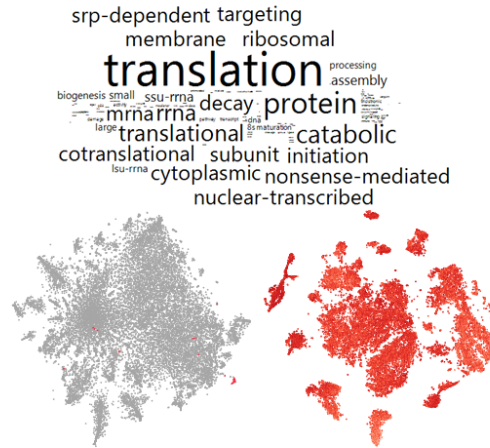
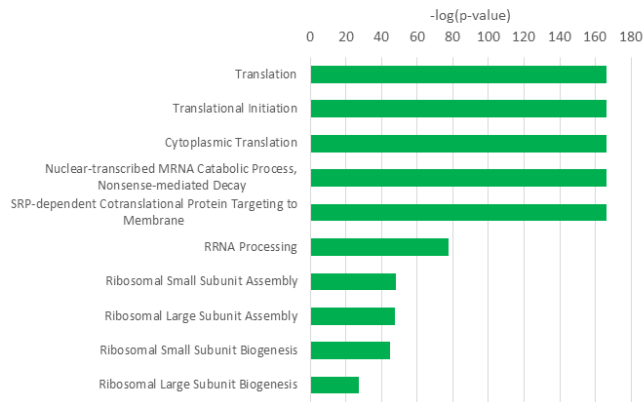
GO:BP enrichment from cluster 39



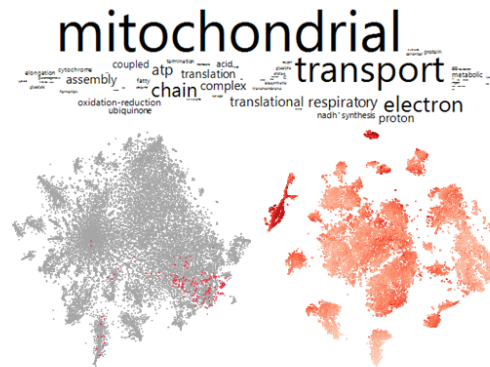
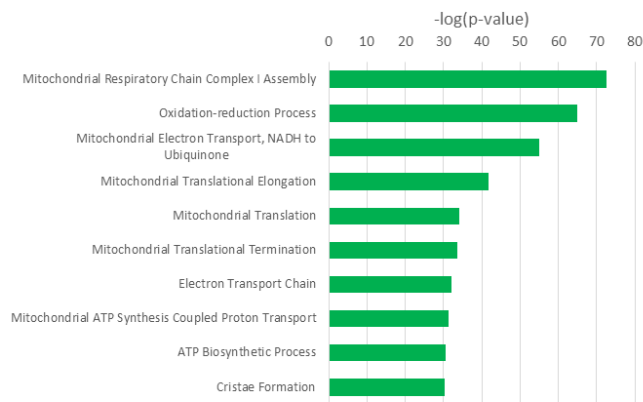
GO:BP enrichment from cluster 40



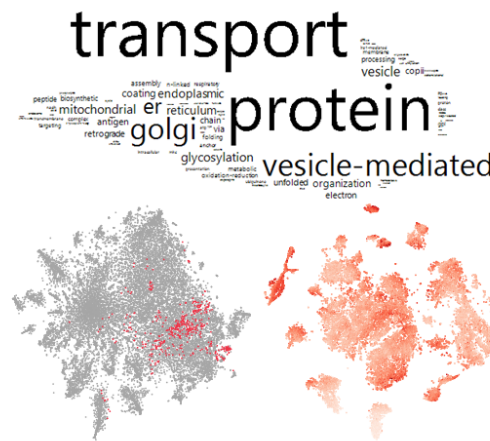
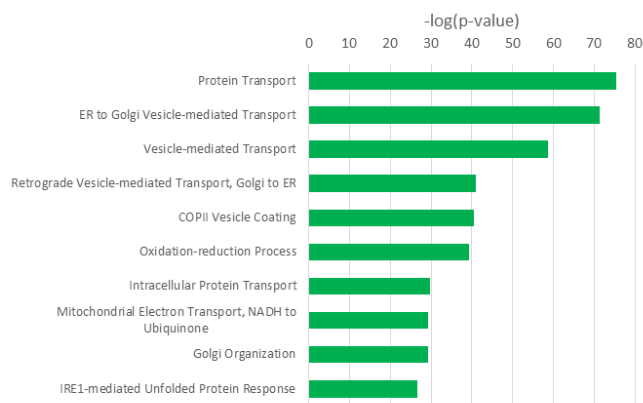
GO:BP enrichment from cluster 41



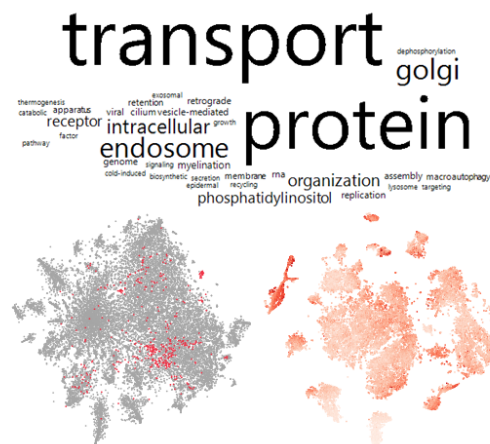
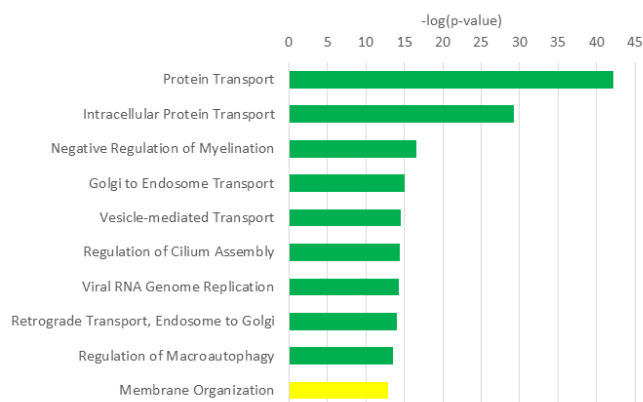
GO:BP enrichment from cluster 42



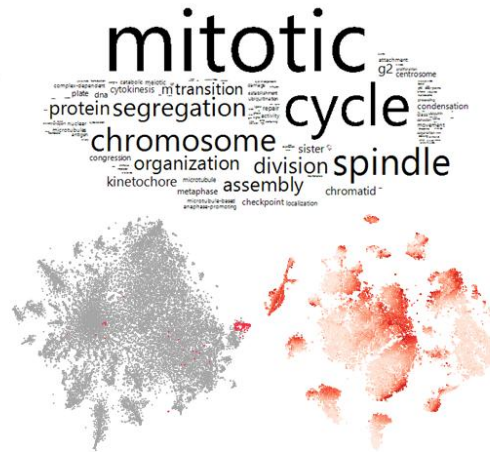
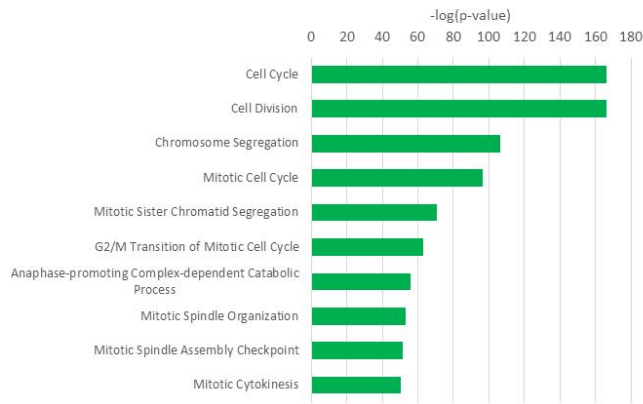
GO:BP enrichment from cluster 43



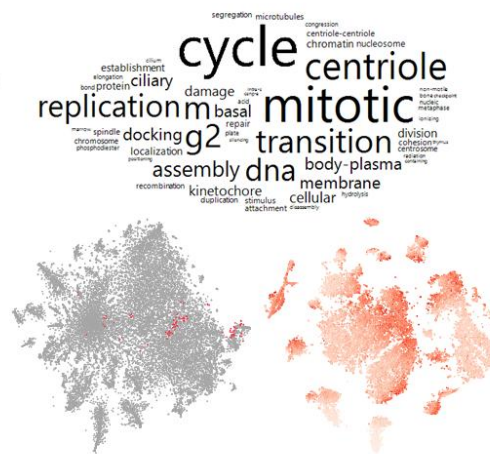
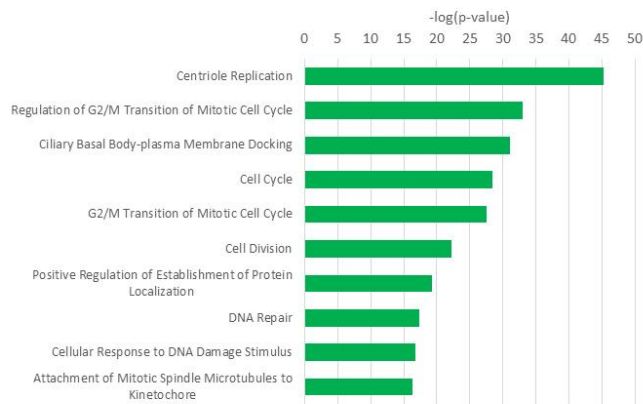
GO:BP enrichment from cluster 44



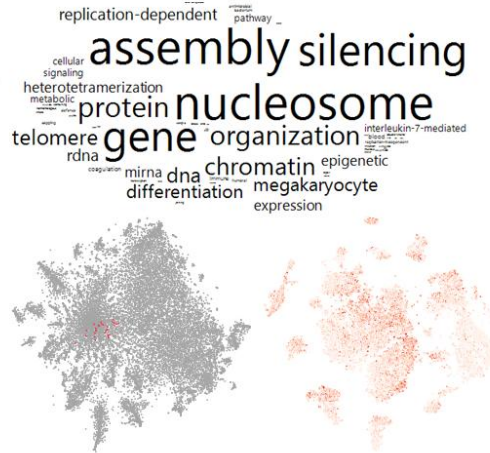
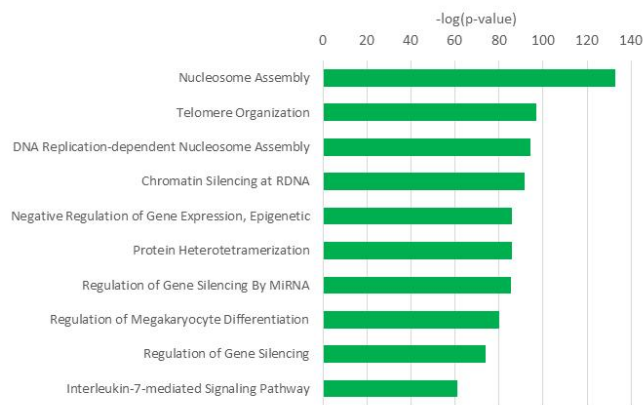
GO:BP enrichment from cluster 45



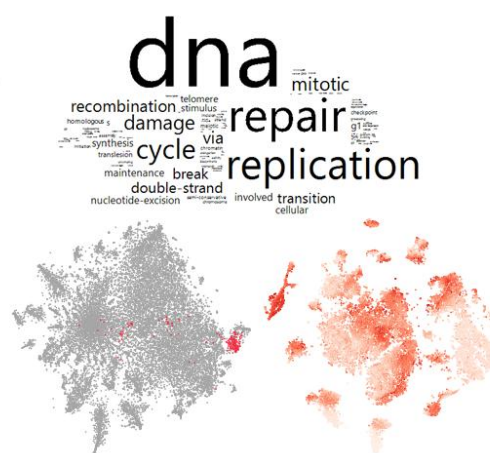
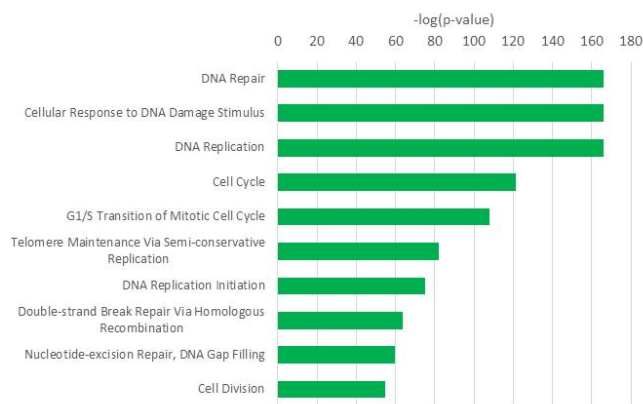
GO:BP enrichment from cluster 46

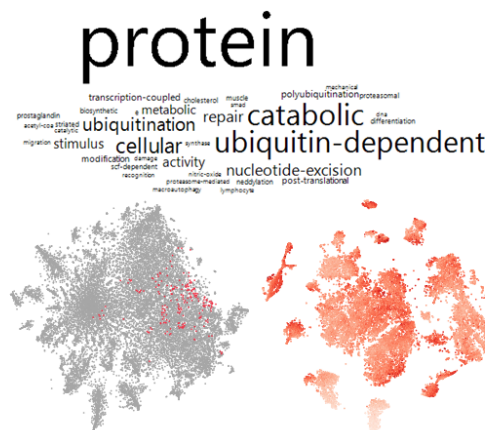
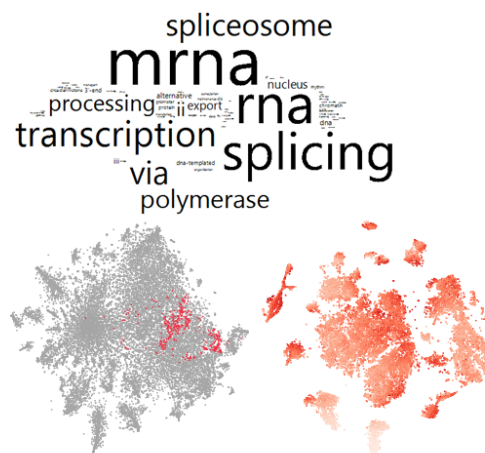
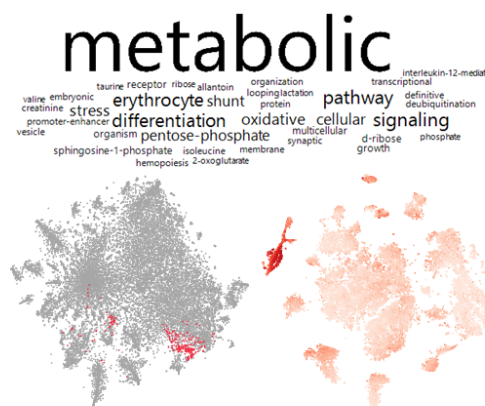
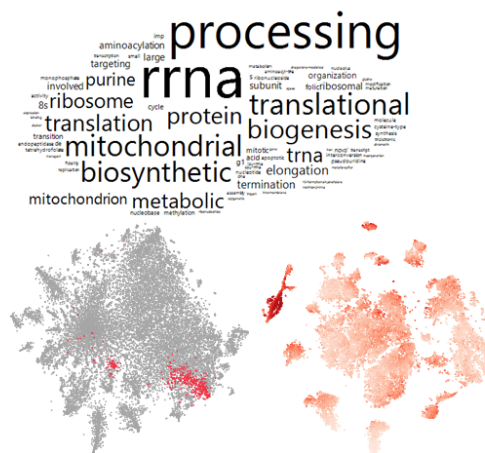


GO:BP enrichment from cluster 47

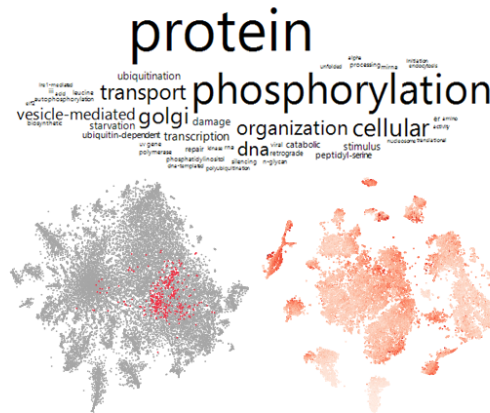
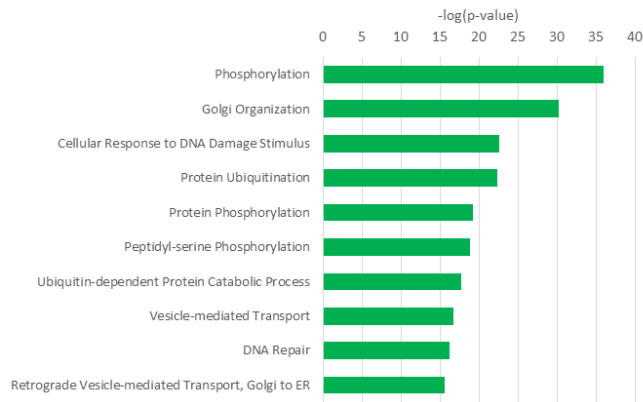


GO:BP enrichment from cluster 48

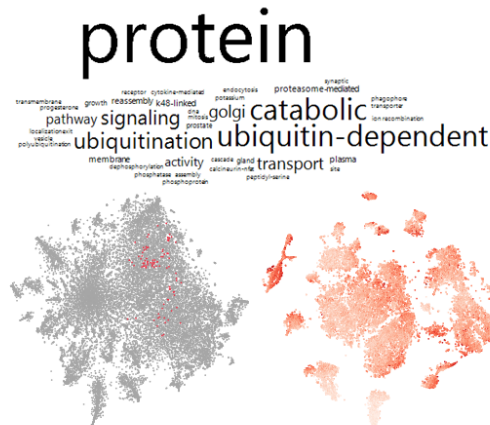
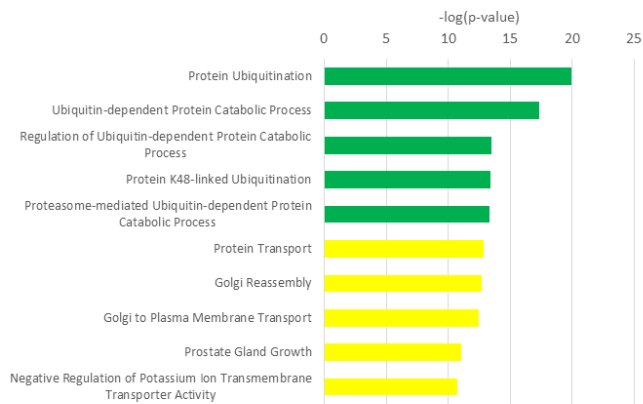




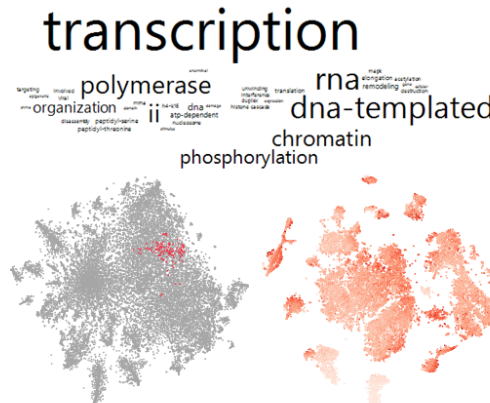
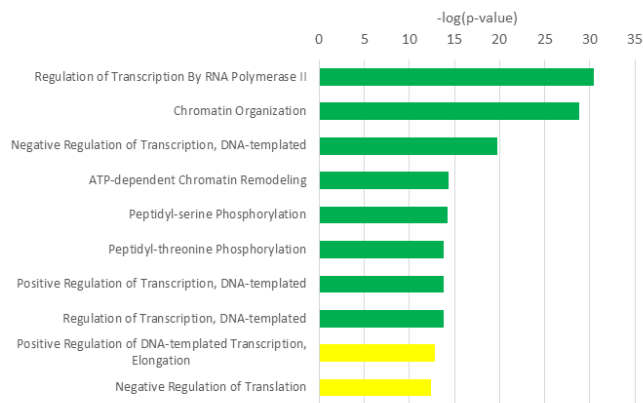
GO:BP enrichment from cluster 53



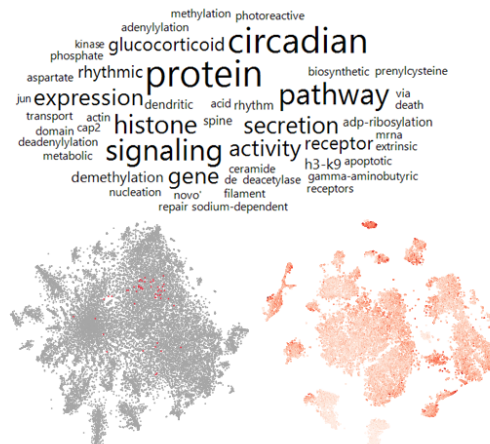
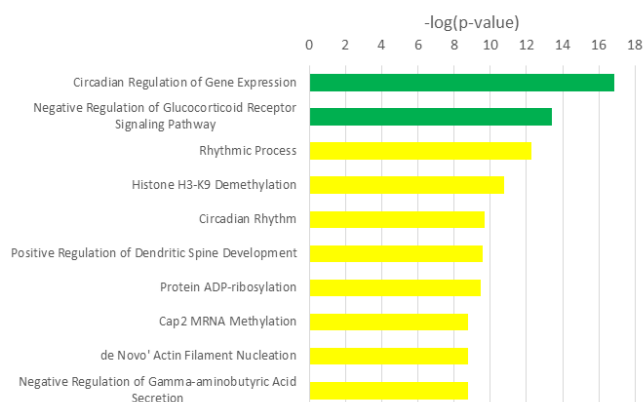
GO:BP enrichment from cluster 54



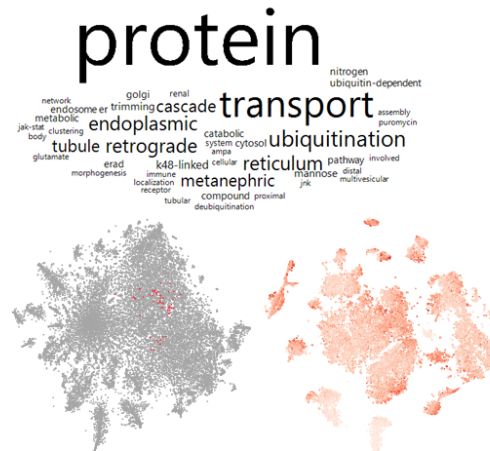
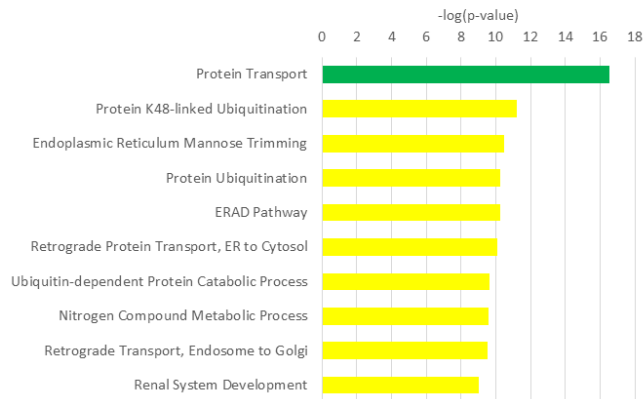
GO:BP enrichment from cluster 55



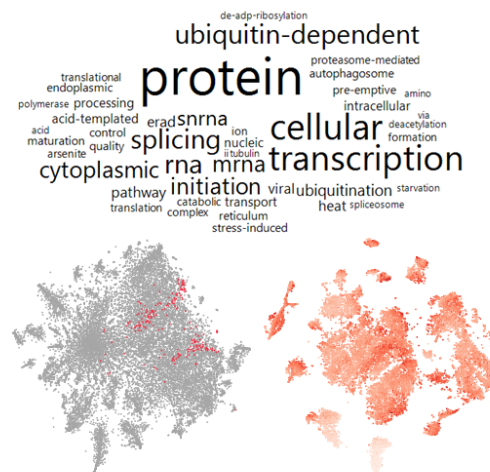
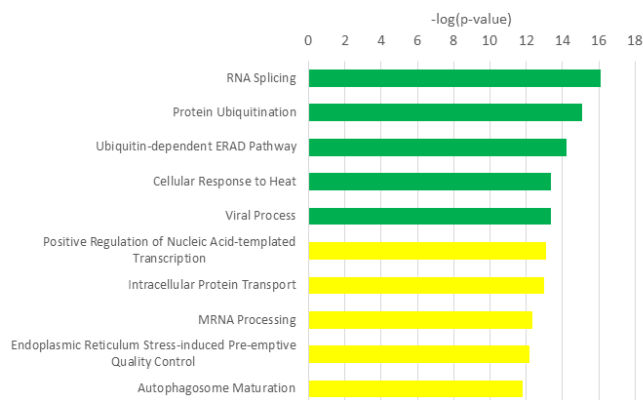
GO:BP enrichment from cluster 56



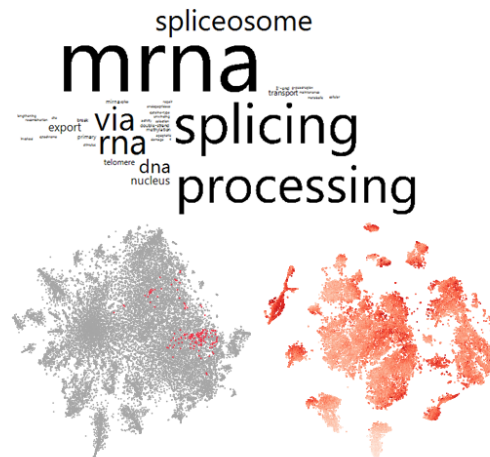
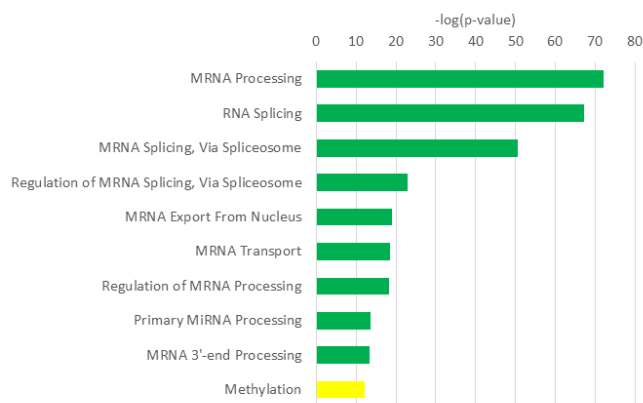
GO:BP enrichment from cluster 57



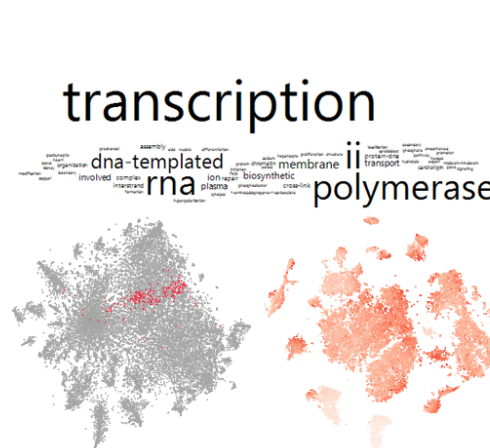
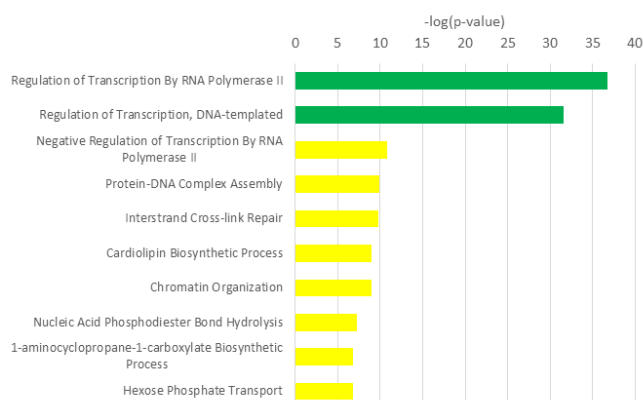
GO:BP enrichment from cluster 58

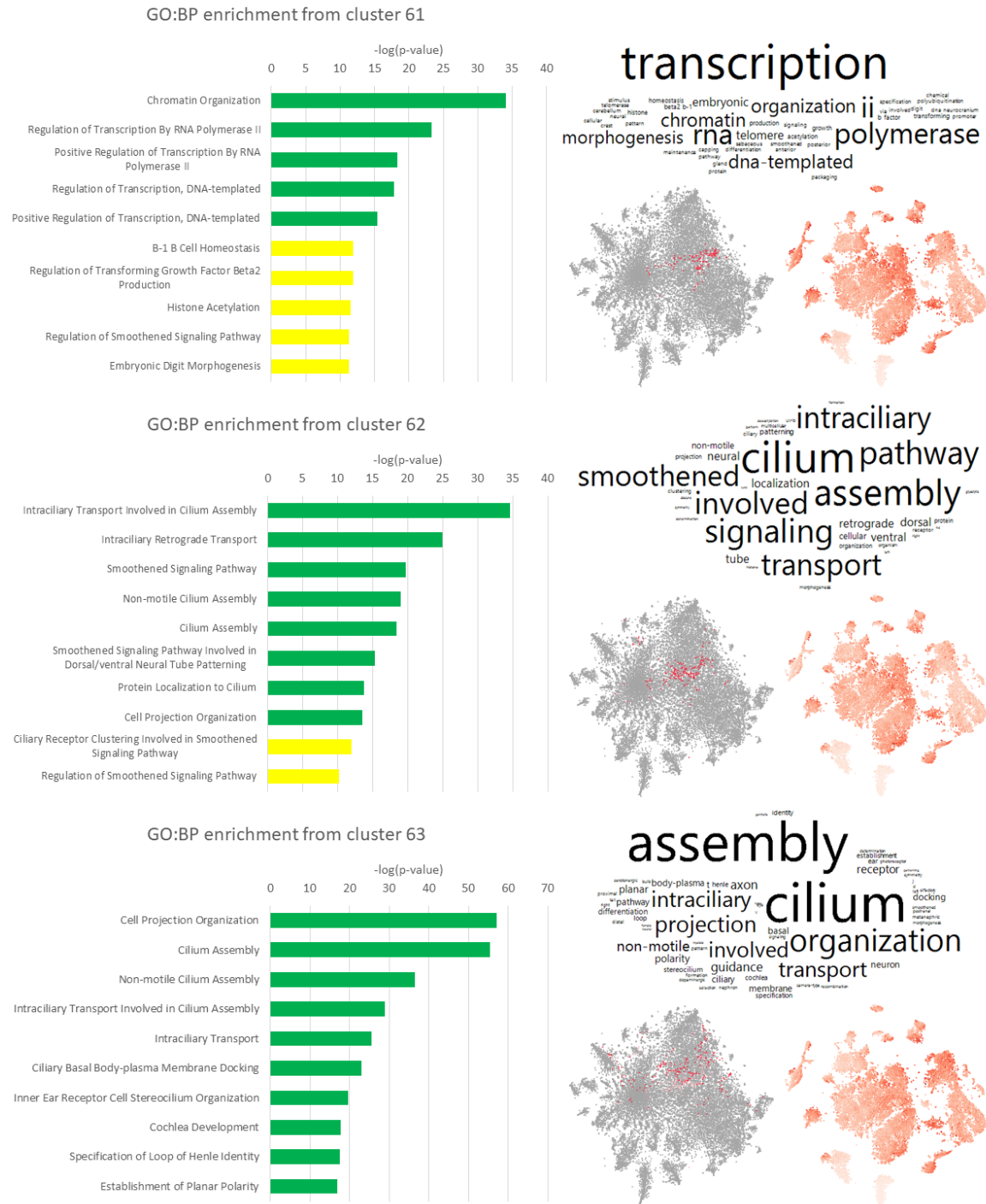


GO:BP enrichment from cluster 59



GO:BP enrichment from cluster 60





Barnum, Kevin J., and Matthew J. O'Connell. 2014. "Cell Cycle Regulation by Checkpoints." *Methods in Molecular Biology* 1170: 29–40.

Butte, A. J., and I. S. Kohane. 2000. "Mutual Information Relevance Networks: Functional Genomic Clustering Using Pairwise Entropy Measurements." *Pacific Symposium on Biocomputing. Pacific Symposium on Biocomputing*, 418–29.

Carter, Scott L., Christian M. Brechbühler, Michael Griffin, and Andrew T. Bond. 2004. "Gene

- Co-Expression Network Topology Provides a Framework for Molecular Characterization of Cellular State." *Bioinformatics* 20 (14): 2242–50.
- Dam, Sipko van, Thomas Craig, and João Pedro de Magalhães. 2015. "GeneFriends: A Human RNA-Seq-Based Gene and Transcript Co-Expression Database." *Nucleic Acids Research* 43 (Database issue): D1124–32.
- Dam, Sipko van, Urmo Vösa, Adriaan van der Graaf, Lude Franke, and João Pedro de Magalhães. 2018. "Gene Co-Expression Analysis for Functional Classification and Gene–disease Predictions." *Briefings in Bioinformatics* 19 (4): 575–92.
- Eisen, M. B., P. T. Spellman, P. O. Brown, and D. Botstein. 1998. "Cluster Analysis and Display of Genome-Wide Expression Patterns." *Proceedings of the National Academy of Sciences of the United States of America* 95 (25): 14863–68.
- Fantes, P. A. 1977. "Control of Cell Size and Cycle Time in *Schizosaccharomyces Pombe*." *Journal of Cell Science* 24 (April): 51–67.
- Hartwell, L. H., and T. A. Weinert. 1989. "Checkpoints: Controls That Ensure the Order of Cell Cycle Events." *Science* 246 (4930): 629–34.
- Hoyt, M. A., L. Totis, and B. T. Roberts. 1991. "S. Cerevisiae Genes Required for Cell Cycle Arrest in Response to Loss of Microtubule Function." *Cell* 66 (3): 507–17.
- Kuntz, Karen, and Matthew J. O'Connell. 2009. "The G(2) DNA Damage Checkpoint: Could This Ancient Regulator Be the Achilles Heel of Cancer?" *Cancer Biology & Therapy* 8 (15): 1433–39.
- Lara-Gonzalez, Pablo, Frederick G. Westhorpe, and Stephen S. Taylor. 2012. "The Spindle Assembly Checkpoint." *Current Biology: CB* 22 (22): R966–80.
- Lemons, Johanna M. S., Xiao-Jiang Feng, Bryson D. Bennett, Aster Legesse-Miller, Elizabeth L. Johnson, Irene Raitman, Elizabeth A. Pollina, Herschel A. Rabitz, Joshua D. Rabinowitz, and Hilary A. Collier. 2010. "Quiescent Fibroblasts Exhibit High Metabolic Activity." *PLoS Biology* 8 (10): e1000514.
- Li, R., and A. W. Murray. 1991. "Feedback Control of Mitosis in Budding Yeast." *Cell* 66 (3): 519–31.
- Luo, Feng, Yunfeng Yang, Jianxin Zhong, Haichun Gao, Latifur Khan, Dorothea K. Thompson, and Jizhong Zhou. 2007. "Constructing Gene Co-Expression Networks and Predicting Functions of Unknown Genes by Random Matrix Theory." *BMC Bioinformatics* 8 (August): 299.
- Mitchison, J. M. 1957. "The Growth of Single Cells. I. *Schizosaccharomyces Pombe*." *Experimental Cell Research* 13 (2): 244–62.
- Mitchison, J. M., and P. Nurse. 1985. "Growth in Cell Length in the Fission Yeast *Schizosaccharomyces Pombe*." *Journal of Cell Science* 75 (April): 357–76.
- O'Connell, Matthew J., and Karlene A. Cimprich. 2005. "G2 Damage Checkpoints: What Is the Turn-On?" *Journal of Cell Science* 118 (Pt 1): 1–6.
- O'Connell, M. J., N. C. Walworth, and A. M. Carr. 2000. "The G2-Phase DNA-Damage Checkpoint." *Trends in Cell Biology* 10 (7): 296–303.
- Ruprecht, Colin, Sebastian Proost, Marcela Hernandez-Coronado, Carlos Ortiz-Ramirez, Daniel Lang, Stefan A. Rensing, Jörg D. Becker, Klaas Vandepoele, and Marek Mutwil. 2017. "Phylogenomic Analysis of Gene Co-Expression Networks Reveals the Evolution of Functional Modules." *The Plant Journal: For Cell and Molecular Biology* 90 (3): 447–65.
- Stuart, Joshua M., Eran Segal, Daphne Koller, and Stuart K. Kim. 2003. "A Gene-Coexpression Network for Global Discovery of Conserved Genetic Modules." *Science* 302 (5643): 249–55.
- Tapia-Alveal, Claudia, Teresa M. Calonge, and Matthew J. O'Connell. 2009. "Regulation of chk1." *Cell Division* 4 (April): 8.
- Weinert, T. A., and L. H. Hartwell. 1988. "The RAD9 Gene Controls the Cell Cycle Response to DNA Damage in *Saccharomyces Cerevisiae*." *Science* 241 (4863): 317–22.

